

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df(data):
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    # Removing of the columns where there is more than 15% of nan
    #df_features = pd.DataFrame(features_all)
    cols_to_check = target_col + features_X
```

```

nan_ratio = df[cols_to_check].isna().mean()
valid_features = nan_ratio[nan_ratio <= 0.15].index
valid_features_list = list(valid_features)

#df = df[valid_features]
# Removing of last np.nan
mask = np.isfinite(df[valid_features]).all(axis=1)
df = df[mask]

cols = list(dict.fromkeys(valid_features_list + #target_col + features_X +
                        feats_to_balance + feats_grouping +
                        feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

new_feats_X = [x for x in features_X if x in valid_features_list]
new_feats_add = [x for x in feats_eng if x in valid_features_list]

data["df"]=df
data["features_X"] = new_feats_X
data["feats_added"] = new_feats_add

column_removed = [x for x in cols_to_check if x not in valid_features_list]
print(f'Column removed because np.isnan > 15%: {column_removed}')

return data

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":

```

```

df["campaign"] = df["datetime"].apply(assign_campaign)
df = df[df["campaign"] != "other"].copy()
df["innova_point"] = df.apply(assign_innova_point, axis=1)
df = df[df["innova_point"] != "no"].copy()
with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r") as f:
    #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
        dict_features = json.load(f)
elif target_col[0] == "i_CO2":
    df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
    df = df[df["campaign"] != "other"].copy()
    df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
    df = df[df["innova_point"] != "no"].copy()
    with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
        dict_features = json.load(f)
else:
    raise ValueError("The target col do not match any existing function")

df["hot"] = df.apply(assign_hot_season,axis=1)
df["umidity_range"] = df.apply(assign_humidity, axis = 1)
#Removing of rabbit 4
df["columnn"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features

```

```

def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}

```

```

        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")

```

Column removed because np.isnan > 15%: ['r_NH3_wl_2h_D2_energy', 'r_NH3_wl_2h_D2_energy_rel', 'r_temperature_wl_2h_D2_energy', 'r_temperature_wl_2h_D2_energy_rel', 'r_umidity_wl_2h_D2_energy', 'r_umidity_wl_2h_D2_energy_rel']

1.2 Metadata

Nombre de lignes df	7639
Nombre de lignes df clean	7639
Nombre de features	47
% de NaN	0.0
Taille dataset entraînement	5347
Taille dataset test	2291
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_2h_D1_energy, r_NH3_wl_2h_A_energy, r_NH3_wl_2h_A_rms, r_NH3_wl_2h_D1_energy_rel, r_NH3_wl_2h_A_energy_rel, r_NH3_wl_12h_D1_energy, r_NH3_wl_12h_D2_energy, r_NH3_wl_12h_D3_energy, r_NH3_wl_12h_A_energy, r_NH3_wl_12h_A_rms, r_NH3_wl_12h_D1_energy_rel, r_NH3_wl_12h_D2_energy_rel, r_NH3_wl_12h_D3_energy_rel, r_NH3_wl_12h_A_energy_rel, r_temperature_wl_2h_D1_energy, r_temperature_wl_2h_A_energy, r_temperature_wl_2h_A_rms, r_temperature_wl_2h_D1_energy_rel, r_temperature_wl_2h_A_energy_rel, r_temperature_wl_12h_D1_energy, r_temperature_wl_12h_D2_energy, r_temperature_wl_12h_D3_energy, r_temperature_wl_12h_A_energy, r_temperature_wl_12h_A_rms, r_temperature_wl_12h_D1_energy_rel, r_temperature_wl_12h_D2_energy_rel, r_temperature_wl_12h_D3_energy_rel, r_temperature_wl_12h_A_energy_rel, r_umidity_wl_2h_D1_energy,

r_umidity_wl_2h_A_energy, r_umidity_wl_2h_A_rms, r_umidity_wl_2h_D1_energy_rel, r_umidity_wl_2h_A_energy_rel, r_umidity_wl_12h_D1_energy, r_umidity_wl_12h_D2_energy, r_umidity_wl_12h_D3_energy, r_umidity_wl_12h_A_energy, r_umidity_wl_12h_A_rms, r_umidity_wl_12h_D1_energy_rel, r_umidity_wl_12h_D2_energy_rel, r_umidity_wl_12h_D3_energy_rel, r_umidity_wl_12h_A_energy_rel

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

The following section has been done with a threshold choosen of : 0.8

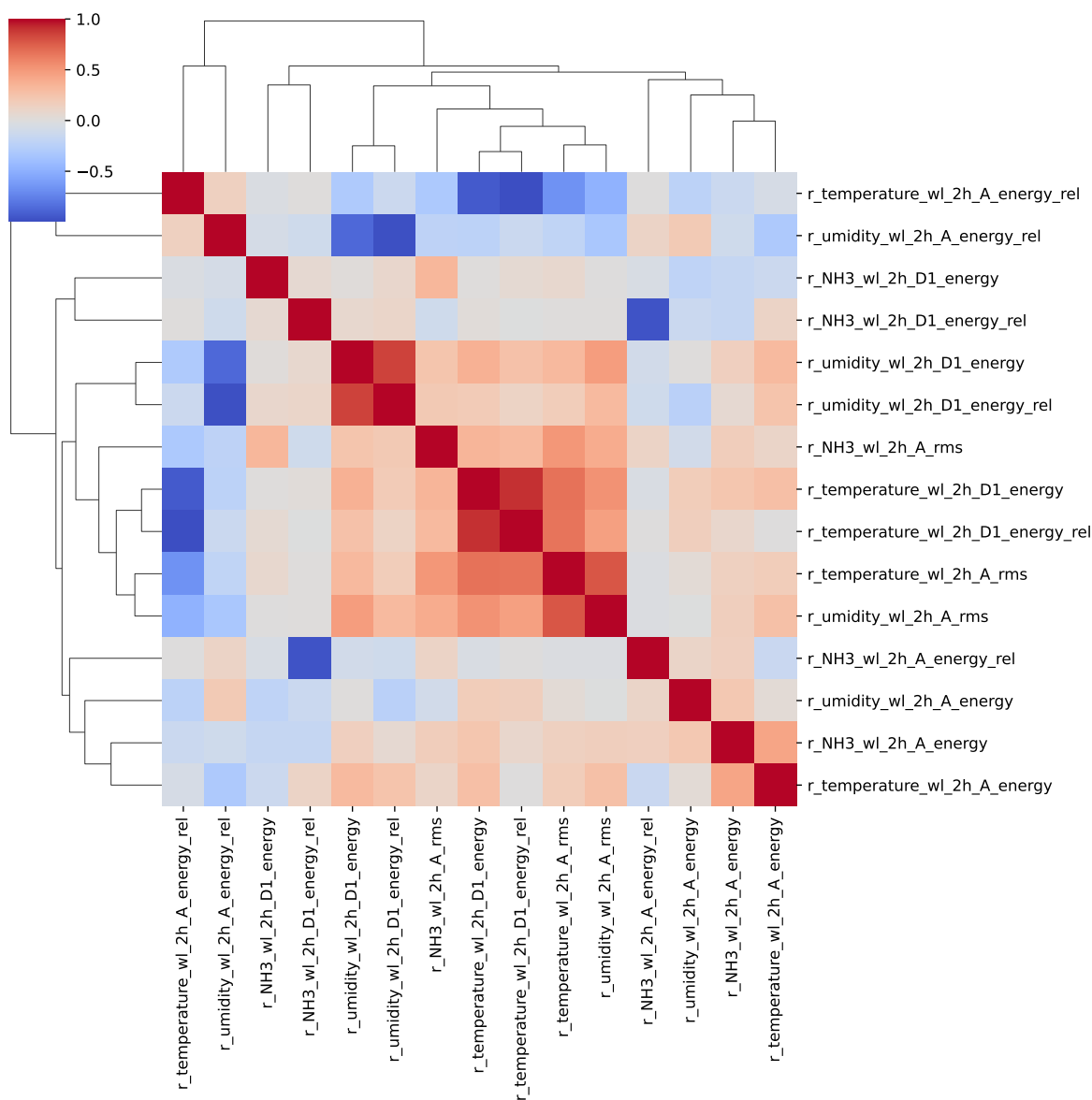
cluster 1: r, , N, H, 3 ; **cluster 2:** r, , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, , u, m, i, d, i, t, y ; **cluster 4:** r, , C, O, 2 ; **cluster 5:** r, __, T, H, I ; **cluster 6:** r_NH3_wl_2h_D1_energy ; **cluster 7:** r_NH3_wl_2h_A_energy ; **cluster 8:** r_NH3_wl_2h_A_rms ; **cluster 9:** r_NH3_wl_2h_A_energy_rel, r_NH3_wl_2h_D1_energy_rel ; **cluster 10:** r_temperature_wl_2h_A_energy_rel, r_temperature_wl_2h_D1_energy, r_temperature_wl_2h_D1_energy_rel ; **cluster 11:** r_temperature_wl_2h_A_energy ; **cluster 12:** r_temperature_wl_2h_A_rms ; **cluster 13:** r_umidity_wl_2h_A_energy_rel, r_umidity_wl_2h_D1_energy_rel, r_umidity_wl_2h_D1_energy ; **cluster 14:** r_umidity_wl_2h_A_energy ; **cluster 15:** r_umidity_wl_2h_A_rms ; **cluster 16:** r_NH3_wl_12h_D1_energy ; **cluster 17:** r_NH3_wl_12h_D2_energy ; **cluster 18:** r_NH3_wl_12h_D3_energy ; **cluster 19:** r_NH3_wl_12h_A_energy ; **cluster 20:** r_NH3_wl_12h_A_rms ; **cluster 21:** r_NH3_wl_12h_A_energy_rel, r_NH3_wl_12h_D2_energy_rel, r_NH3_wl_12h_D1_energy_rel ; **cluster 22:** r_NH3_wl_12h_D3_energy_rel ; **cluster 23:** r_temperature_wl_12h_D2_energy_rel, r_temperature_wl_12h_D1_energy_rel, r_temperature_wl_12h_A_energy_rel, r_temperature_wl_12h_D1_energy, r_temperature_wl_12h_D2_energy ; **cluster 24:** r_temperature_wl_12h_D3_energy, r_temperature_wl_12h_D3_energy_rel ; **cluster 25:** r_temperature_wl_12h_A_energy ; **cluster 26:** r_temperature_wl_12h_A_rms ; **cluster 27:** r_umidity_wl_12h_D2_energy, r_umidity_wl_12h_D1_energy, r_umidity_wl_12h_D1_energy_rel, r_umidity_wl_12h_A_energy_rel, r_umidity_wl_12h_D2_energy_rel ; **cluster 28:** r_umidity_wl_12h_D3_energy,

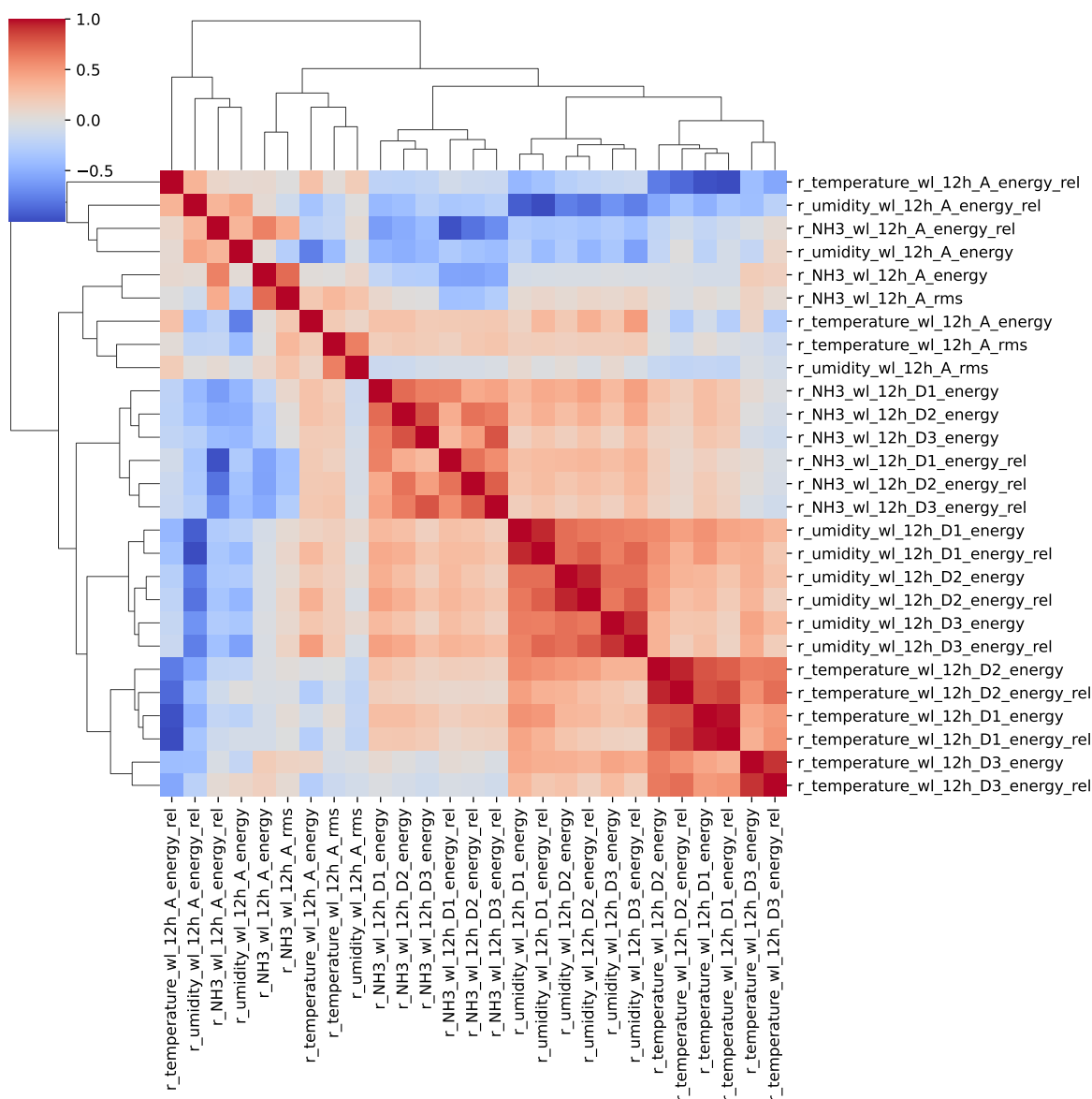
r_umidity_wl_12h_D3_energy_rel ; **cluster 29:** r_umidity_wl_12h_A_energy ; **cluster 30:** r_umidity_wl_12h_A_rms ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_2h_D1_energy, r_NH3_wl_2h_A_energy, r_NH3_wl_2h_A_rms, r_NH3_wl_2h_A_energy_rel, r_temperature_wl_2h_A_energy_rel, r_temperature_wl_2h_A_energy, r_temperature_wl_2h_A_rms, r_umidity_wl_2h_A_energy_rel, r_umidity_wl_2h_A_energy, r_umidity_wl_2h_A_rms, r_NH3_wl_12h_D1_energy, r_NH3_wl_12h_D2_energy, r_NH3_wl_12h_D3_energy, r_NH3_wl_12h_A_energy, r_NH3_wl_12h_A_rms, r_NH3_wl_12h_A_energy_rel, r_NH3_wl_12h_D3_energy_rel, r_temperature_wl_12h_D2_energy_rel, r_temperature_wl_12h_D3_energy, r_temperature_wl_12h_A_energy, r_temperature_wl_12h_A_rms, r_umidity_wl_12h_D2_energy, r_umidity_wl_12h_D3_energy, r_umidity_wl_12h_A_energy, r_umidity_wl_12h_A_rms





3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_2h_D1_energy, r_NH3_wl_2h_A_energy, r_NH3_wl_2h_A_rms, r_NH3_wl_2h_A_energy_rel,

r_temperature_wl_2h_A_energy_rel, r_temperature_wl_2h_A_energy, r_temperature_wl_2h_A_rms, r_umidity_wl_2h_A_energy_rel, r_umidity_wl_2h_A_energy, r_umidity_wl_2h_A_rms, r_NH3_wl_12h_D1_energy, r_NH3_wl_12h_D2_energy, r_NH3_wl_12h_D3_energy, r_NH3_wl_12h_A_energy, r_NH3_wl_12h_A_rms, r_NH3_wl_12h_A_energy_rel, r_NH3_wl_12h_D3_energy_rel, r_temperature_wl_12h_D2_energy_rel, r_temperature_wl_12h_D3_energy, r_temperature_wl_12h_A_energy, r_temperature_wl_12h_A_rms, r_umidity_wl_12h_D2_energy, r_umidity_wl_12h_D3_energy, r_umidity_wl_12h_A_energy, r_umidity_wl_12h_A_rms

Len(Features_X): 30

Config Gridsearch:{‘show’: True, ‘search_type’: ‘optuna’, ‘model_name’: None, ‘model’: None, ‘param_grid’: None, ‘best_param’: None, ‘cv_func’: ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), ‘save_models’: True, ‘sumup-table’: True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

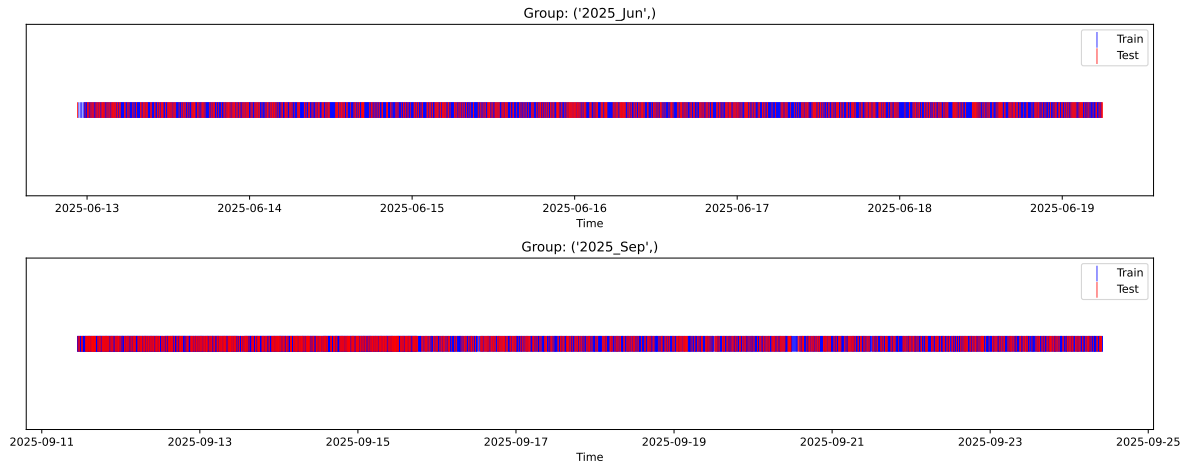
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_{train} vs X_{test}

External Split: Train: 5347 samples Test: 2292 samples



3.3.2 Internal CV Folds

Fold 1/5

Train: 3742 samples Test: 1605 samples Excluded: 0 samples Window exclusion: 0h



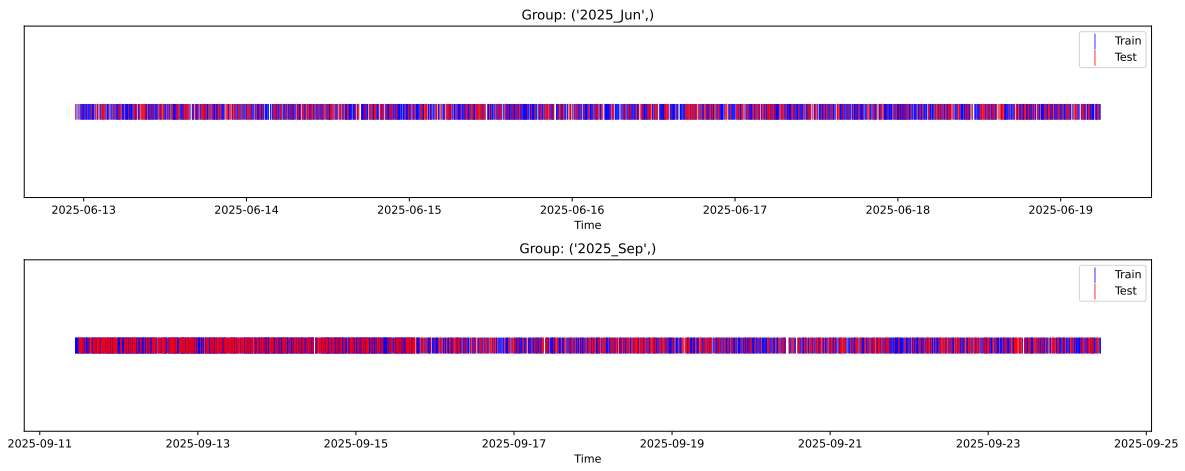
Fold 2/5

Train: 3742 samples Test: 1605 samples Excluded: 0 samples Window exclusion: 0h



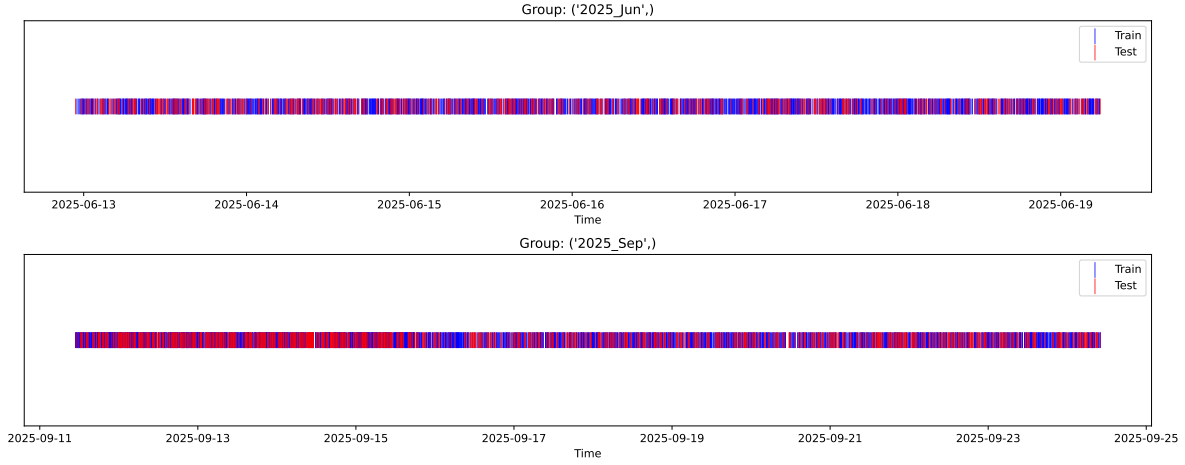
Fold 3/5

Train: 3742 samples Test: 1605 samples Excluded: 0 samples Window exclusion: 0h



Fold 4/5

Train: 3742 samples Test: 1605 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3742 samples Test: 1605 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.334, std=0.792, min=0.959, max=13.845
 y_{test} : mean=2.361, std=0.816, min=0.973, max=7.779

Fold 0 — y_{test} : mean=2.34, std=0.83, y_{train} : mean=2.33, std=0.77

Fold 1 — y_{test} : mean=2.31, std=0.76, y_{train} : mean=2.35, std=0.80

Fold 2 — y_{test} : mean=2.37, std=0.80, y_{train} : mean=2.32, std=0.79

Fold 3 — y_test: mean=2.35, std=0.79, y_train: mean=2.33, std=0.79

Fold 4 — y_test: mean=2.34, std=0.76, y_train: mean=2.33, std=0.80

train scores CV: [0.57802094 0.54219308 0.5527182 0.54545337 0.54681618]

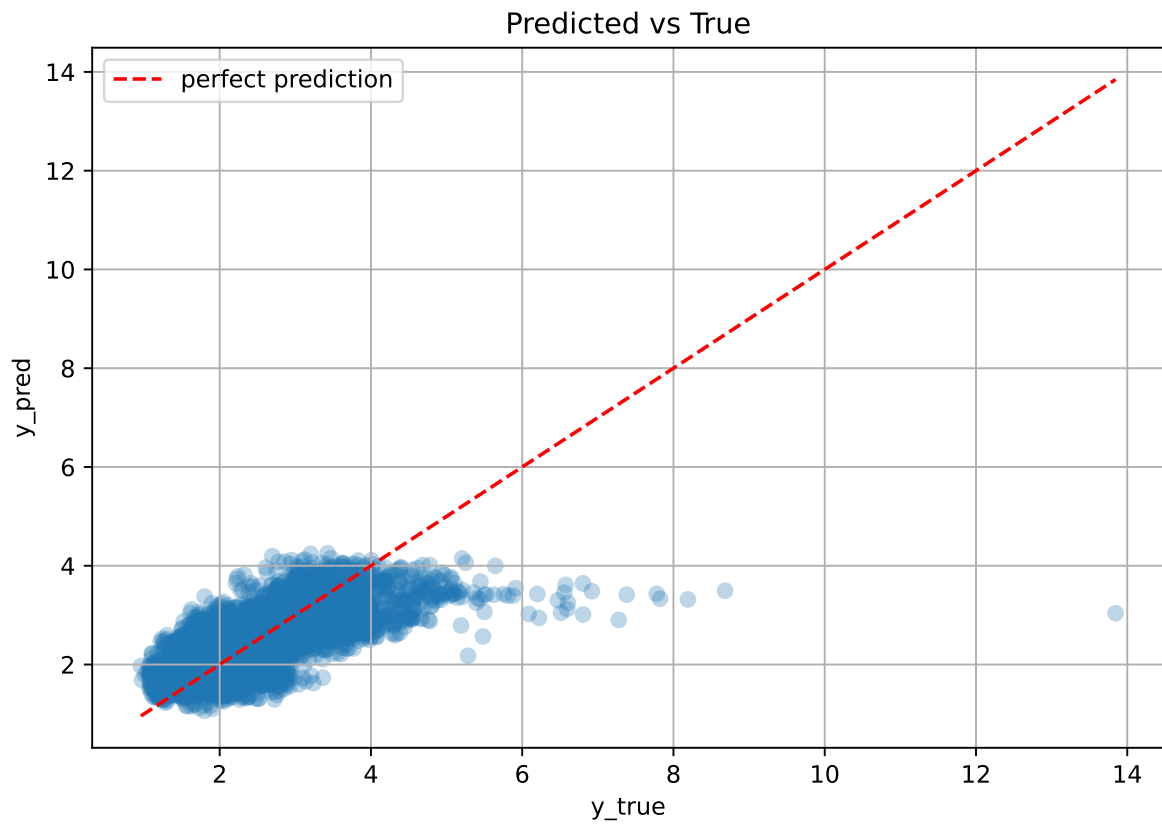
test scores CV: [0.50107971 0.57675573 0.55122131 0.56872698 0.56577172]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.553	0.5527	0.0271	0.5583	0.5598	0.0071	1.0128	1.0006

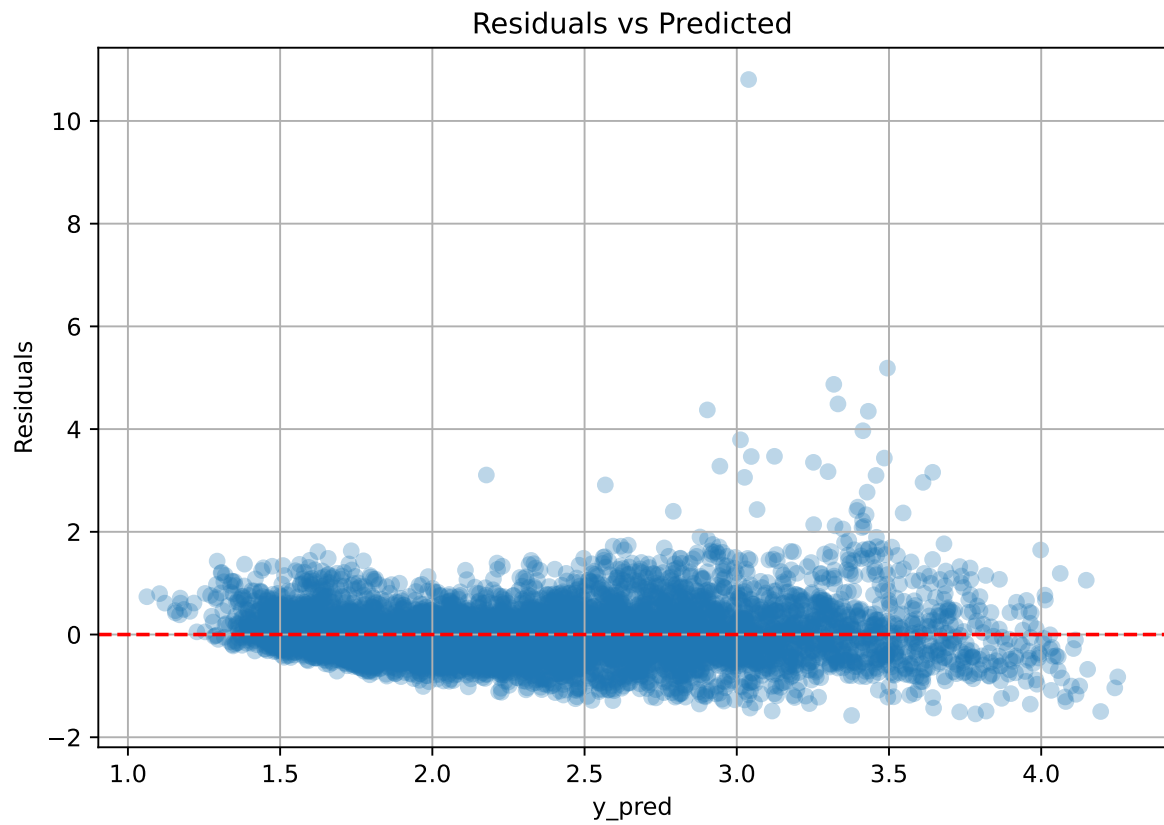
Coefficients:

coef_r_CO2 = -0.0617, coef_r_NH3 = 0.2553, coef_r_NH3_wl_12h_A_energy = -0.1470, coef_r_NH3_wl_12h_A_energy_rel = -0.0044, coef_r_NH3_wl_12h_A_rms = 0.0341, coef_r_NH3_wl_12h_D1_energy = 0.0513, coef_r_NH3_wl_12h_D2_energy = 0.0708, coef_r_NH3_wl_12h_D3_energy = -0.0141, coef_r_NH3_wl_12h_D3_energy_rel = -0.0343, coef_r_NH3_wl_2h_A_energy = -0.2159, coef_r_NH3_wl_2h_A_energy_rel = 0.0071, coef_r_NH3_wl_2h_A_rms = -0.0005, coef_r_NH3_wl_2h_D1_energy = -0.0160, coef_r_THI = 0.4053, coef_r_temperature = -0.3837, coef_r_temperature_wl_12h_A_energy = -0.0324, coef_r_temperature_wl_12h_A_rms = 0.0001, coef_r_temperature_wl_12h_D2_energy_rel = 0.0177, coef_r_temperature_wl_12h_D3_energy = -0.0593, coef_r_temperature_wl_2h_A_energy = -0.0560, coef_r_temperature_wl_2h_A_energy_rel = -0.0281, coef_r_temperature_wl_2h_A_rms = 0.1486, coef_r_umidity = -0.5873, coef_r_umidity_wl_12h_A_energy = 0.0832, coef_r_umidity_wl_12h_A_rms = -0.1053, coef_r_umidity_wl_12h_D2_energy = -0.0340, coef_r_umidity_wl_12h_D3_energy = 0.0408, coef_r_umidity_wl_2h_A_energy = 0.1385, coef_r_umidity_wl_2h_A_energy_rel = -0.0390, coef_r_umidity_wl_2h_A_rms = 0.0270, intercept = 2.3341,

3.4.2 Scatter



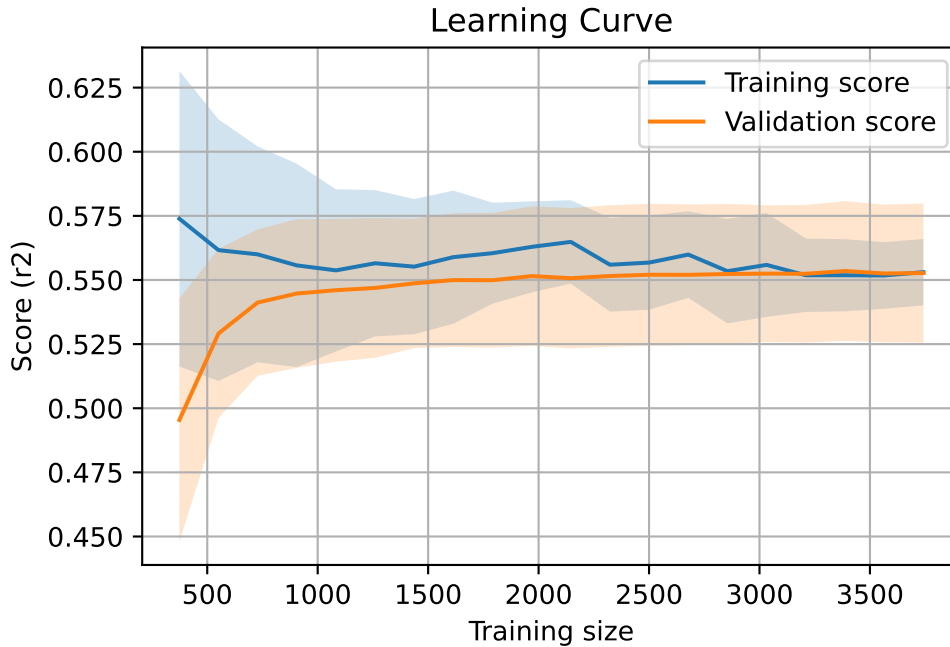
3.4.3 Scatter residuals



3.4.4 Learning curve

`Len(group)` : 7639, `Len(training)` : 5348,

`Len(training_real)` : 3744, `Len(test_of_training)` : 1604, `Len(test)` : 2291



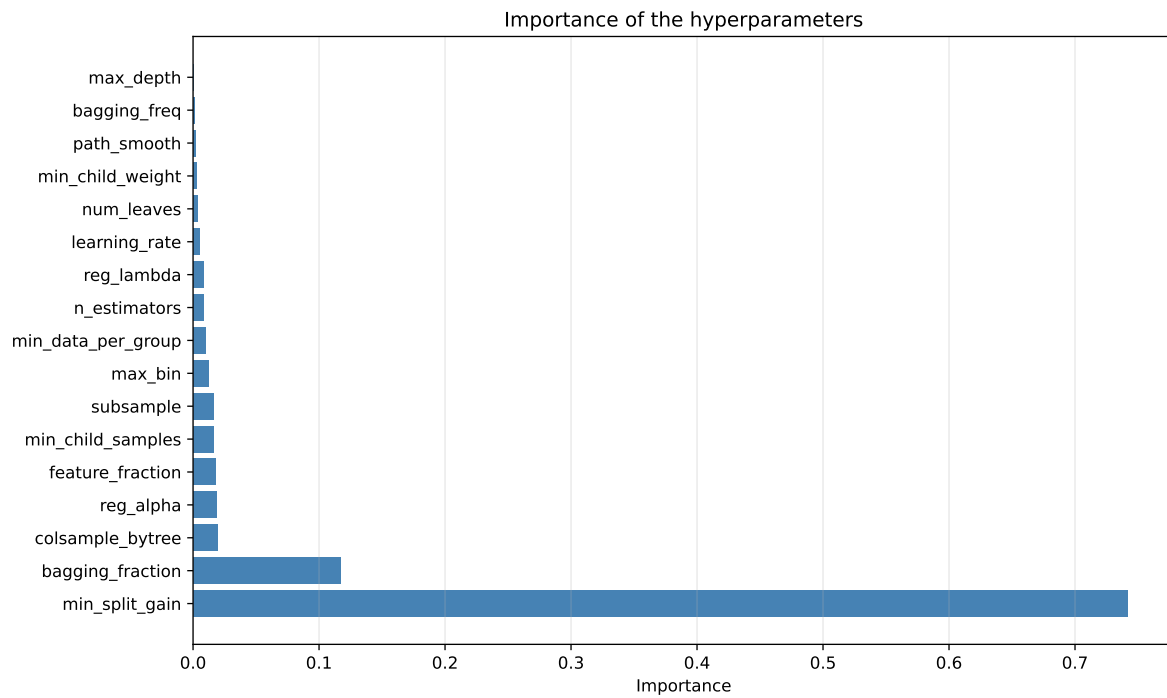
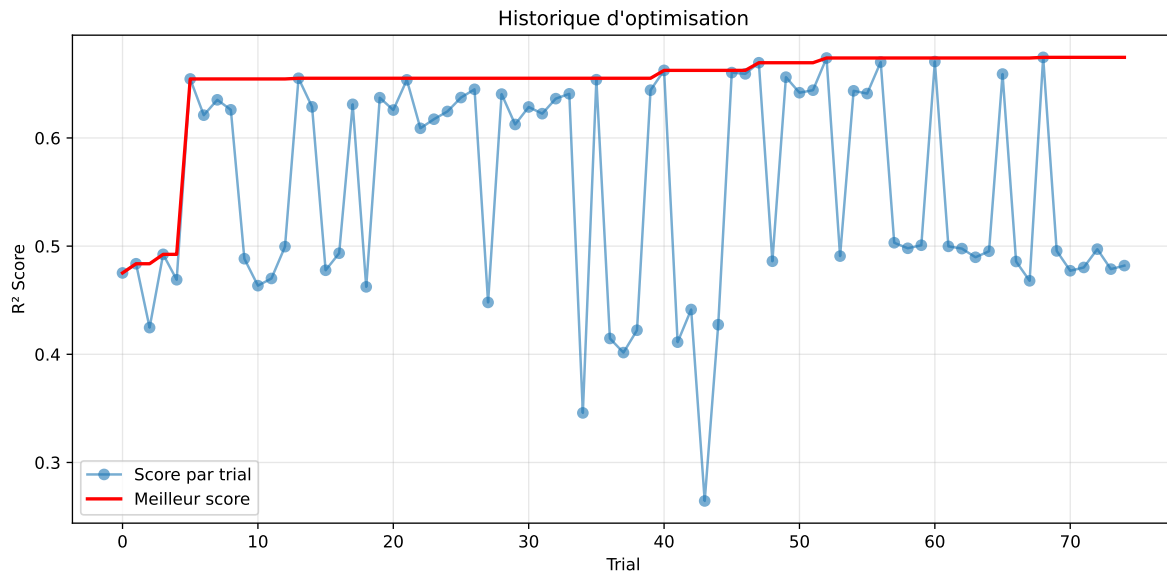
3.5 Model : LGBM

3.5.1 Gridsearch

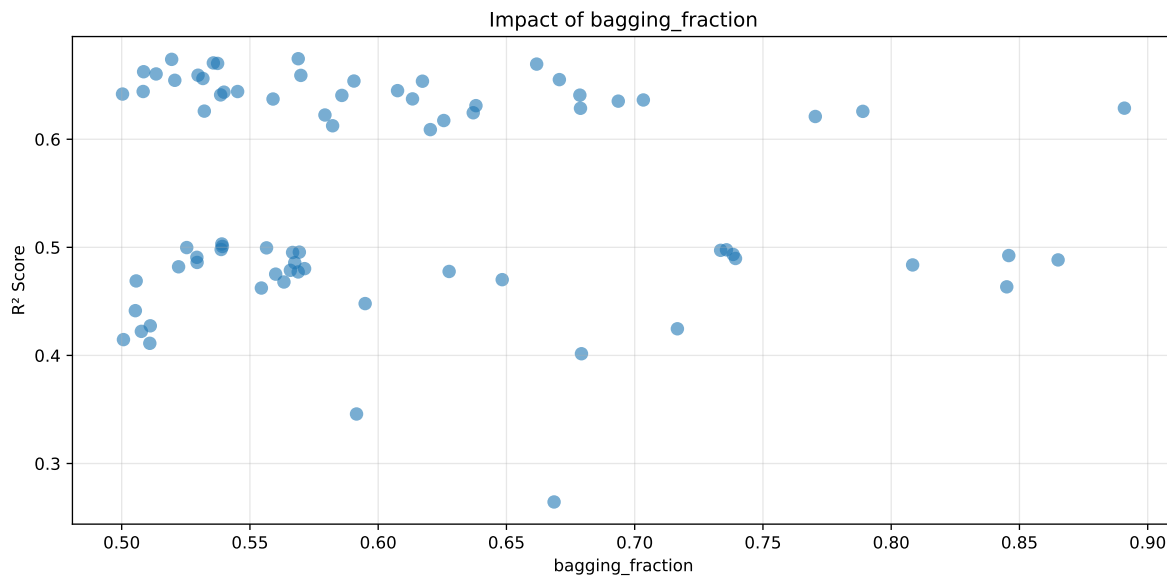
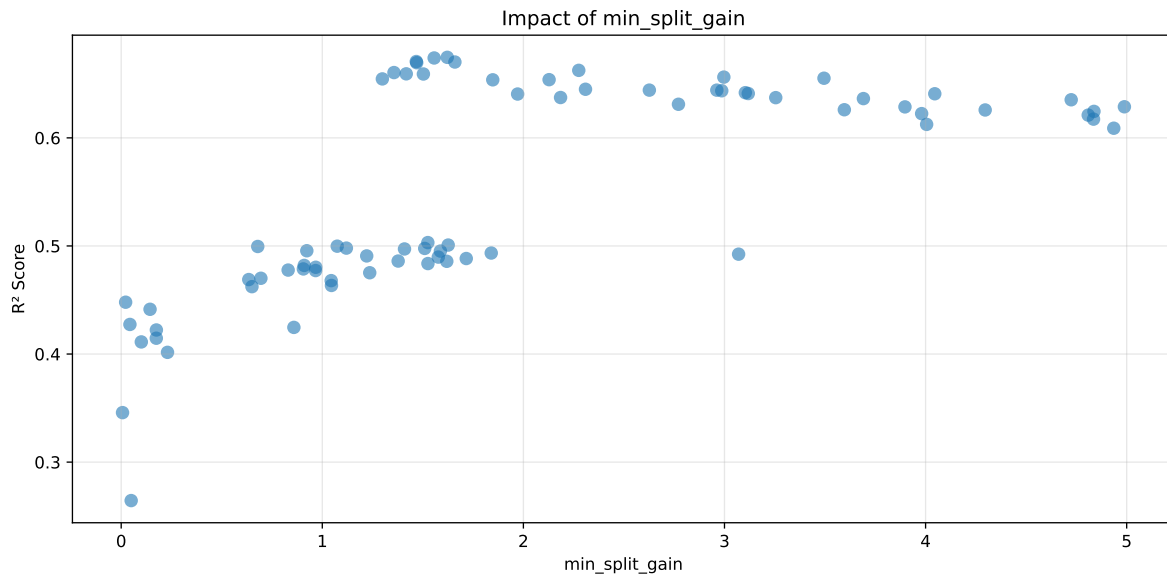
Distribution y_{train} vs y_{test} : y_{train} : mean=2.334, std=0.792, min=0.959, max=13.845
 y_{test} : mean=2.361, std=0.816, min=0.973, max=7.779

===== TOP Importance FEATURES =====

feature	importance
r_CO2	380
r_temperature_wl_12h_A_energy	282
r_umidity	278
r_temperature_wl_2h_A_rms	171
r_umidity_wl_12h_A_rms	167
r_temperature_wl_12h_A_rms	155
r_NH3	150
r_temperature	148
r_NH3_wl_12h_D1_energy	145
r_NH3_wl_12h_D3_energy_rel	137
r_THI	124
r_NH3_wl_12h_A_energy	116
r_NH3_wl_2h_A_energy	107
r_NH3_wl_2h_D1_energy	99
r_NH3_wl_12h_D3_energy	92
r_temperature_wl_2h_A_energy_rel	83
r_NH3_wl_12h_D2_energy	81
r_umidity_wl_2h_A_energy	77
r_umidity_wl_12h_D3_energy	75
r_temperature_wl_12h_D3_energy	66
r_umidity_wl_2h_A_energy_rel	64
r_temperature_wl_12h_D2_energy_rel	59
r_NH3_wl_2h_A_rms	57
r_umidity_wl_12h_A_energy	49
r_NH3_wl_12h_A_rms	46
r_NH3_wl_12h_A_energy_rel	38
r_temperature_wl_2h_A_energy	37
r_umidity_wl_12h_D2_energy	36
r_umidity_wl_2h_A_rms	30
r_NH3_wl_2h_A_energy_rel	18

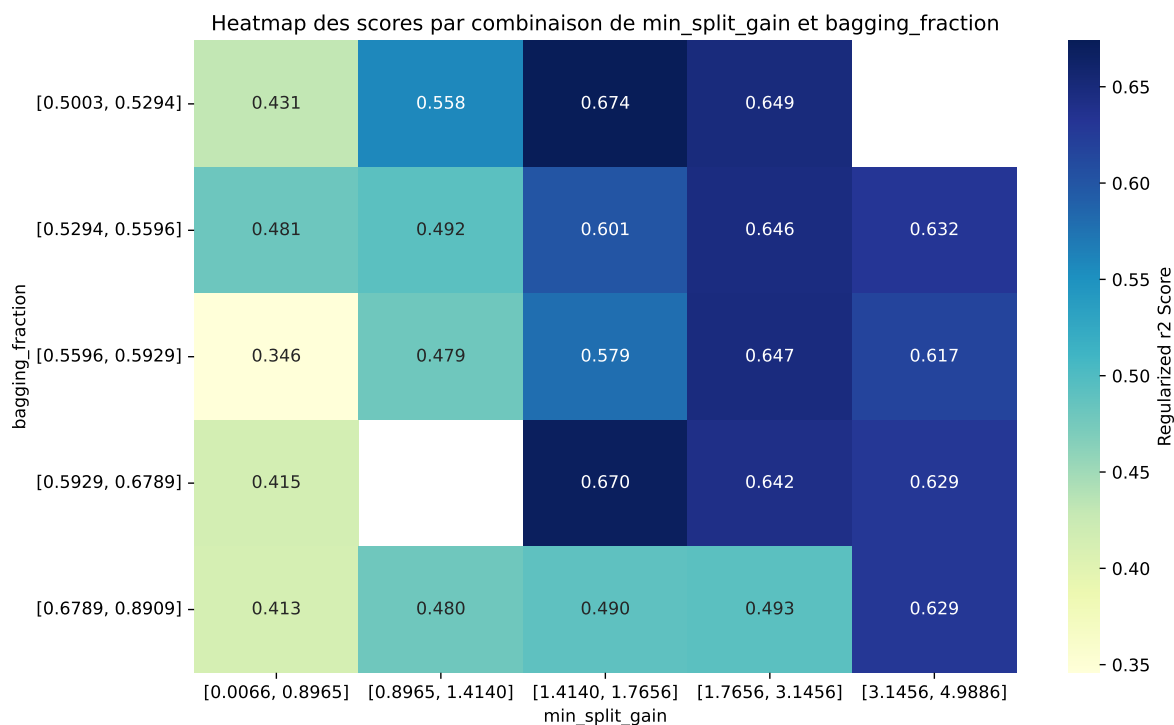


Paramètres avec >5% d'importance : ['min_split_gain', 'bagging_fraction']



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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.707	0.6745	0.6833	0.0385	0.6745	0.7103	0.0357	1.053	1.0481
2	0.7063	0.6739	0.6833	0.0382	0.6739	0.7025	0.0286	1.0425	1.0481
3	0.7016	0.6706	0.6795	0.0398	0.6706	0.7067	0.0361	1.0538	1.0462
4	0.7034	0.6702	0.6795	0.0386	0.6702	0.7035	0.0334	1.0498	1.0496
5	0.7029	0.6695	0.679	0.0389	0.6695	0.7079	0.0384	1.0573	1.0498
6	0.6899	0.6625	0.6712	0.0368	0.6625	0.6895	0.027	1.0408	1.0414
7	0.6859	0.6604	0.6678	0.0387	0.6604	0.7003	0.0399	1.0604	1.0386
8	0.6841	0.6592	0.6656	0.0373	0.6592	0.7009	0.0416	1.0631	1.0377
9	0.6875	0.6591	0.6669	0.0378	0.6591	0.7003	0.0412	1.0625	1.0432
10	0.6838	0.6562	0.6661	0.0396	0.6562	0.6865	0.0303	1.0462	1.042
---	---	---	---	---	---	---	---	---	---
75	0.8634	0.7636	0.7829	0.0392	0.2644	0.8078	0.0442	1.0579	1.1307

Hyperparameters:

Rank 1: {'reg__n_estimators': 600, 'reg__max_depth': 5, 'reg__learning_rate': 0.015703472615455842, 'reg__num_leaves': 30, 'reg__subsample': 0.6161975300653267, 'reg__colsample_bytree': 0.7102651352426557, 'reg__min_child_samples': 46, 'reg__reg_alpha': 1.5990756398346613, 'reg__reg_lambda': 37.36641443593545, 'reg__min_split_gain': 1.621539846086606, 'reg__path_smooth': 3.5512154423866775, 'reg__min_child_weight':

0.03245751204959584, 'reg__feature_fraction': 0.6921637795220482, 'reg__bagging_fraction': 0.568832491312766, 'reg__bagging_freq': 6, 'reg__max_bin': 99, 'reg__min_data_per_group': 92}

Rank 2: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.02017451162812159, 'reg__num_leaves': 34, 'reg__subsample': 0.6860759666910484, 'reg__colsample_bytree': 0.8006616385913066, 'reg__min_child_samples': 46, 'reg__reg_alpha': 2.4054709264926757, 'reg__reg_lambda': 7.783575322979441, 'reg__min_split_gain': 1.5563294467241842, 'reg__path_smooth': 4.077587243339485, 'reg__min_child_weight': 0.07750667861108244, 'reg__feature_fraction': 0.7125191176111874, 'reg__bagging_fraction': 0.5194848354607199, 'reg__bagging_freq': 5, 'reg__max_bin': 77, 'reg__min_data_per_group': 65}

Rank 3: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.015315759906607806, 'reg__num_leaves': 41, 'reg__subsample': 0.6178592279891744, 'reg__colsample_bytree': 0.8244242144423556, 'reg__min_child_samples': 46, 'reg__reg_alpha': 1.6303543700849392, 'reg__reg_lambda': 38.027922751079785, 'reg__min_split_gain': 1.4673974127618221, 'reg__path_smooth': 3.4245355071811403, 'reg__min_child_weight': 9.434164713792383, 'reg__feature_fraction': 0.6819735419529779, 'reg__bagging_fraction': 0.5357337019431518, 'reg__bagging_freq': 6, 'reg__max_bin': 90, 'reg__min_data_per_group': 94}

Rank 4: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.022483880033853886, 'reg__num_leaves': 35, 'reg__subsample': 0.6356015646512472, 'reg__colsample_bytree': 0.7949253071429486, 'reg__min_child_samples': 43, 'reg__reg_alpha': 2.387688903164185, 'reg__reg_lambda': 7.425356889735927, 'reg__min_split_gain': 1.6594216893237101, 'reg__path_smooth': 4.013611153035434, 'reg__min_child_weight': 0.09657537847353696, 'reg__feature_fraction': 0.7148503769436387, 'reg__bagging_fraction': 0.537397497538527, 'reg__bagging_freq': 5, 'reg__max_bin': 89, 'reg__min_data_per_group': 64}

Rank 5: {'reg__n_estimators': 700, 'reg__max_depth': 5, 'reg__learning_rate': 0.021379054226762017, 'reg__num_leaves': 33, 'reg__subsample': 0.6222979651579216, 'reg__colsample_bytree': 0.8365513111759206, 'reg__min_child_samples': 41, 'reg__reg_alpha': 4.223239444948769, 'reg__reg_lambda': 40.565398464782305, 'reg__min_split_gain': 1.4694708834546009, 'reg__path_smooth': 3.8317688321687657, 'reg__min_child_weight': 0.007864143774029005, 'reg__feature_fraction': 0.7543440963231407, 'reg__bagging_fraction': 0.6617813732661778, 'reg__bagging_freq': 5, 'reg__max_bin': 97, 'reg__min_data_per_group': 68}

Rank 6: {'reg__n_estimators': 500, 'reg__max_depth': 4, 'reg__learning_rate': 0.03599283992236343, 'reg__num_leaves': 39, 'reg__subsample': 0.7679193545211666, 'reg__colsample_bytree': 0.8393583787370682, 'reg__min_child_samples': 43, 'reg__reg_alpha': 4.235174344249714, 'reg__reg_lambda': 4.427040478716624, 'reg__min_split_gain': 2.2753794615875687, 'reg__path_smooth': 3.78315039731753, 'reg__min_child_weight':

0.003336676878167015, 'reg__feature_fraction': 0.5913396732806089, 'reg__bagging_fraction': 0.5085511993193342, 'reg__bagging_freq': 1, 'reg__max_bin': 68, 'reg__min_data_per_group': 103}

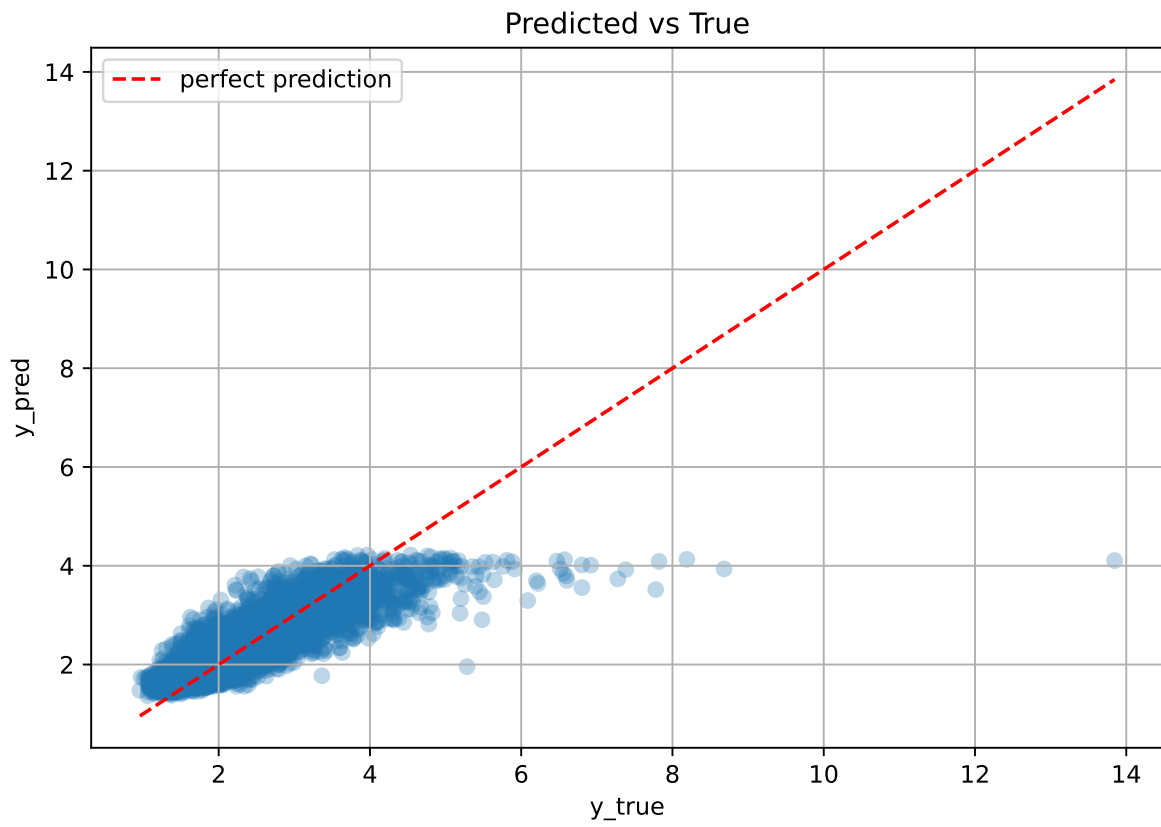
Rank 7: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.021382137878889783, 'reg__num_leaves': 34, 'reg__subsample': 0.6308853654060809, 'reg__colsample_bytree': 0.8430880577602099, 'reg__min_child_samples': 43, 'reg__reg_alpha': 4.131663909327404, 'reg__reg_lambda': 40.17674485591572, 'reg__min_split_gain': 1.3564569131399615, 'reg__path_smooth': 4.000219326020361, 'reg__min_child_weight': 0.007504064813224009, 'reg__feature_fraction': 0.5719092642028867, 'reg__bagging_fraction': 0.5134205566600614, 'reg__bagging_freq': 5, 'reg__max_bin': 64, 'reg__min_data_per_group': 66}

Rank 8: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.02202943885759682, 'reg__num_leaves': 35, 'reg__subsample': 0.6340667139190005, 'reg__colsample_bytree': 0.8435518181574675, 'reg__min_child_samples': 42, 'reg__reg_alpha': 4.23766465773401, 'reg__reg_lambda': 40.953104550622626, 'reg__min_split_gain': 1.417510207530604, 'reg__path_smooth': 3.942356256816841, 'reg__min_child_weight': 0.00792650364124002, 'reg__feature_fraction': 0.5862609630606819, 'reg__bagging_fraction': 0.5297646679088865, 'reg__bagging_freq': 5, 'reg__max_bin': 64, 'reg__min_data_per_group': 66}

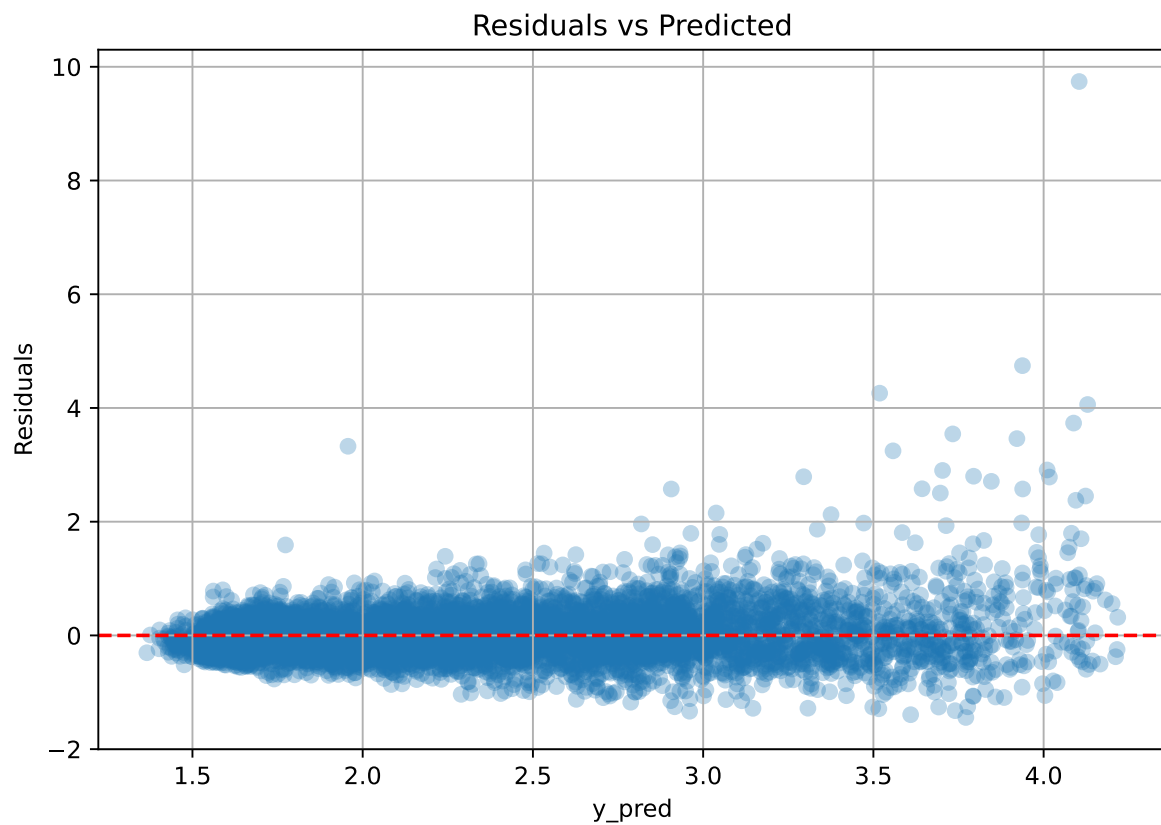
Rank 9: {'reg__n_estimators': 600, 'reg__max_depth': 5, 'reg__learning_rate': 0.01708619286057529, 'reg__num_leaves': 40, 'reg__subsample': 0.6234594322034442, 'reg__colsample_bytree': 0.8196053564848418, 'reg__min_child_samples': 47, 'reg__reg_alpha': 5.288943248869929, 'reg__reg_lambda': 36.373128152245, 'reg__min_split_gain': 1.5027193678004478, 'reg__path_smooth': 3.5256306399042794, 'reg__min_child_weight': 0.14838433719483082, 'reg__feature_fraction': 0.6840986933699701, 'reg__bagging_fraction': 0.5698478356493438, 'reg__bagging_freq': 6, 'reg__max_bin': 96, 'reg__min_data_per_group': 93}

Rank 10: {'reg__n_estimators': 200, 'reg__max_depth': 4, 'reg__learning_rate': 0.05580498719090766, 'reg__num_leaves': 34, 'reg__subsample': 0.7895945999216517, 'reg__colsample_bytree': 0.7913346100789824, 'reg__min_child_samples': 45, 'reg__reg_alpha': 2.0597075696841625, 'reg__reg_lambda': 6.5563110138333585, 'reg__min_split_gain': 2.9968169133780918, 'reg__path_smooth': 4.1375148498162, 'reg__min_child_weight': 0.07672473181751385, 'reg__feature_fraction': 0.6960188398960928, 'reg__bagging_fraction': 0.5316701456611322, 'reg__bagging_freq': 1, 'reg__max_bin': 68, 'reg__min_data_per_group': 66}

3.5.2 Scatter



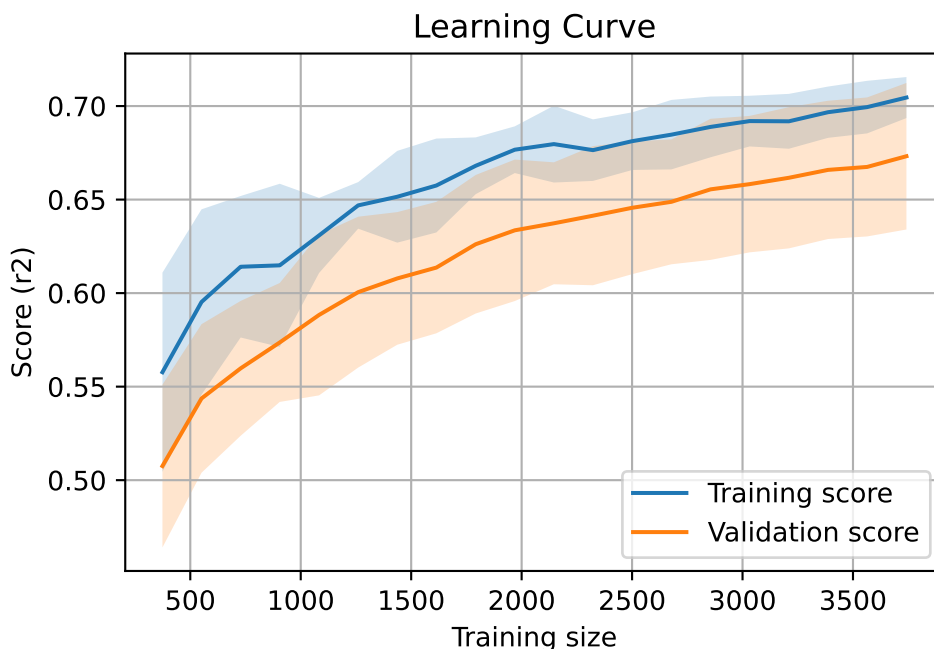
3.5.3 Scatter residuals



3.5.4 Learning curve

`Len(group)` : 7639, `Len(training)` : 5348,

`Len(training_real)` : 3744, `Len(test_of_training)` : 1604, `Len(test)` : 2291



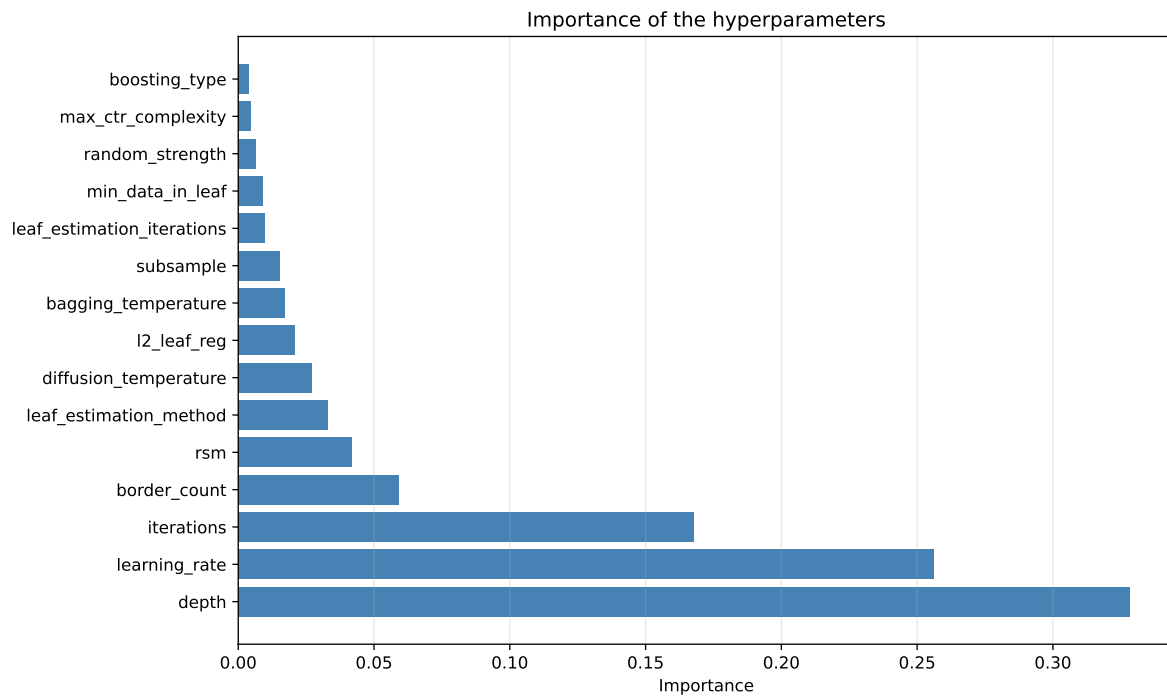
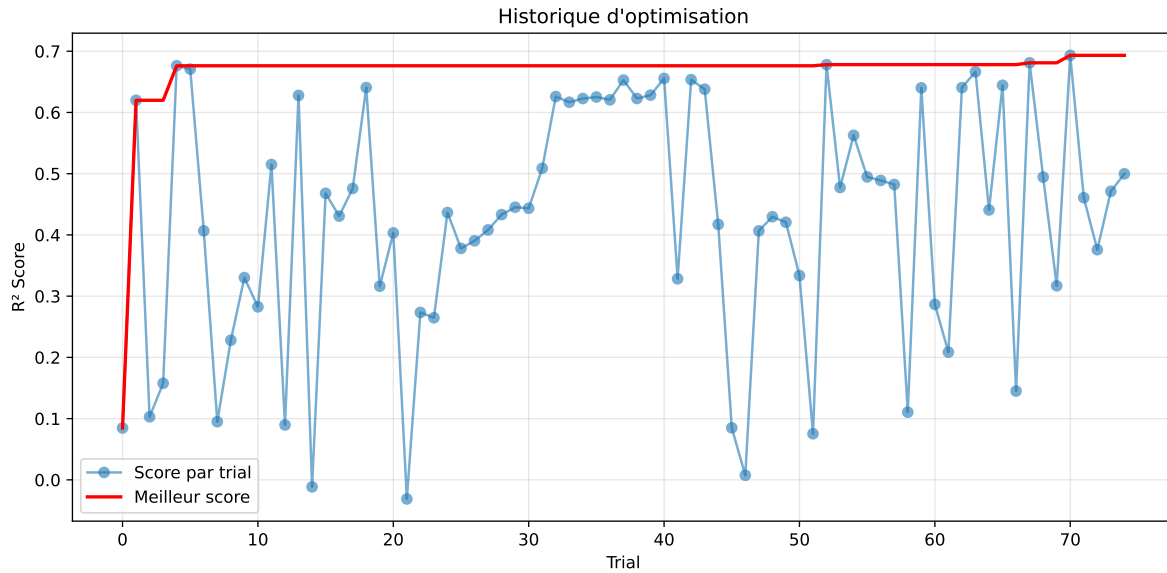
3.6 Model : CatBoost

3.6.1 Gridsearch

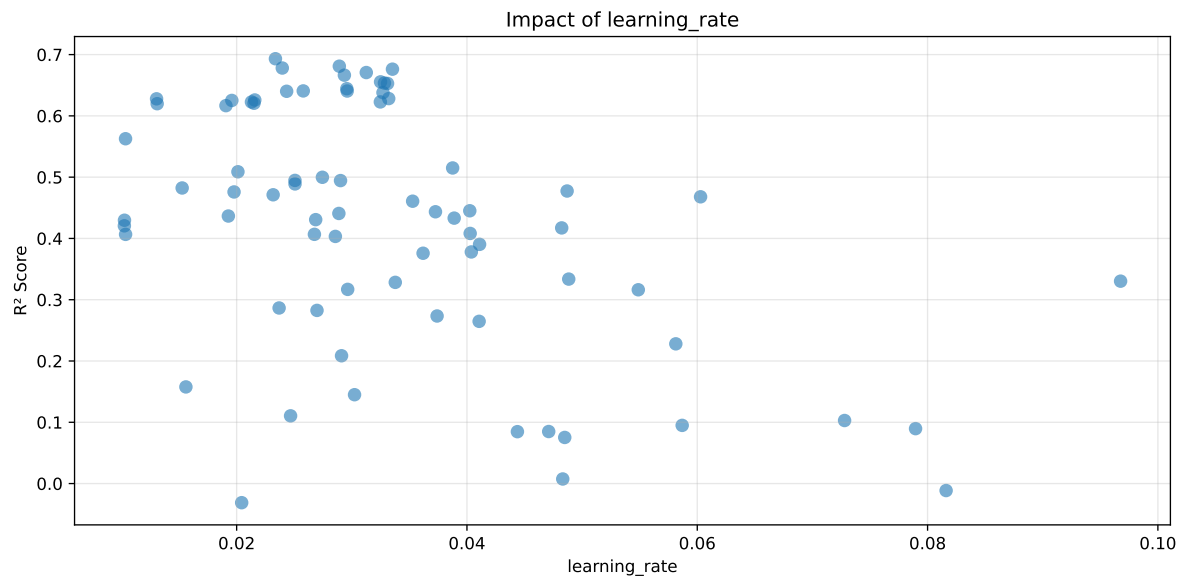
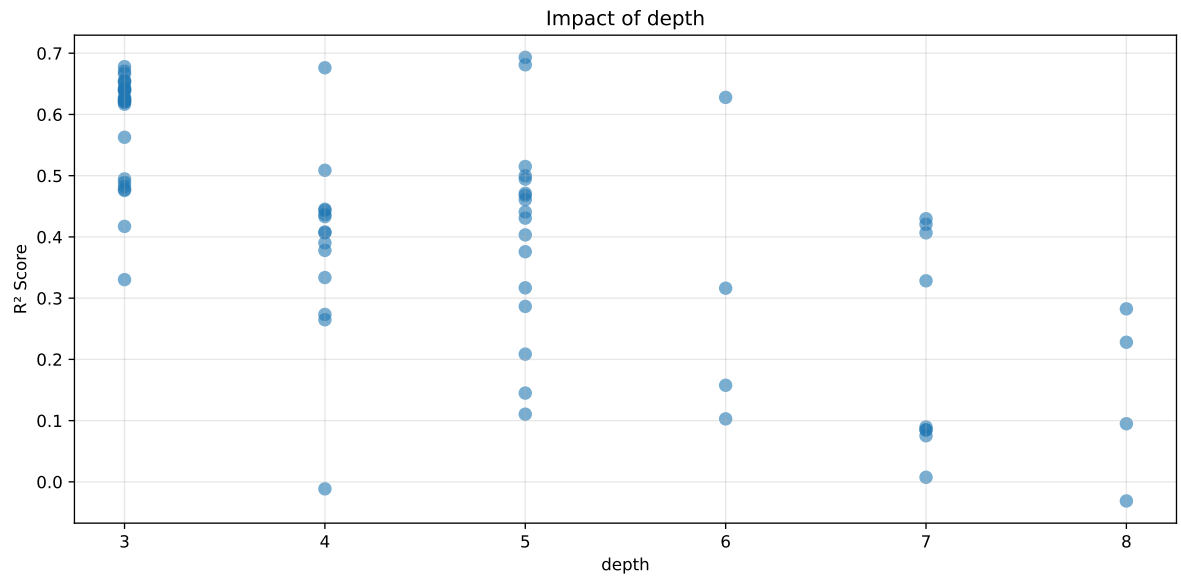
Distribution y_{train} vs y_{test} : y_{train} : mean=2.334, std=0.792, min=0.959, max=13.845
 y_{test} : mean=2.361, std=0.816, min=0.973, max=7.779

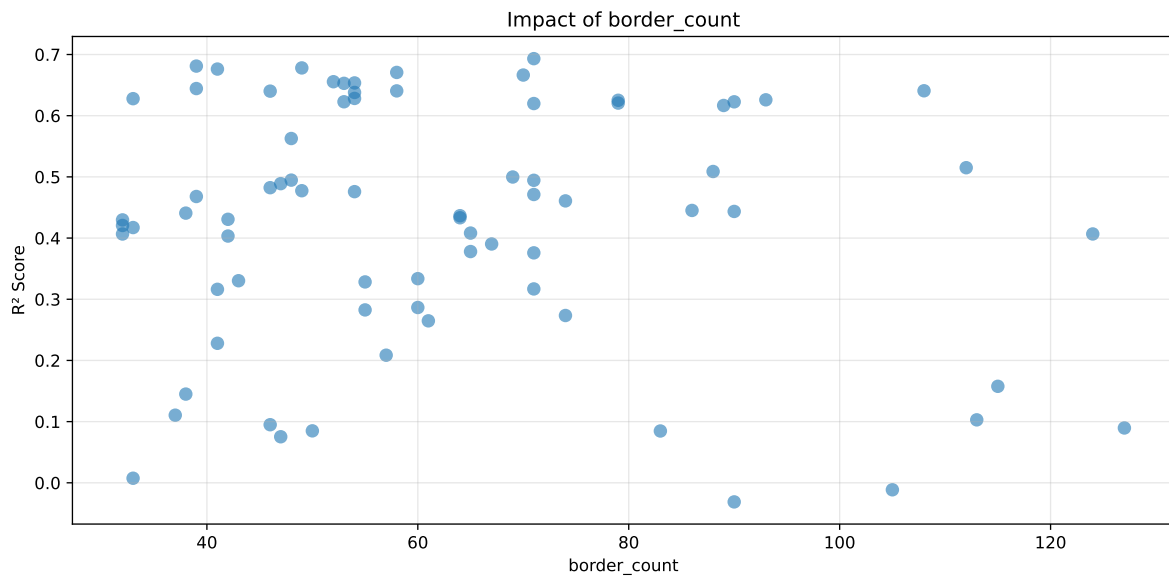
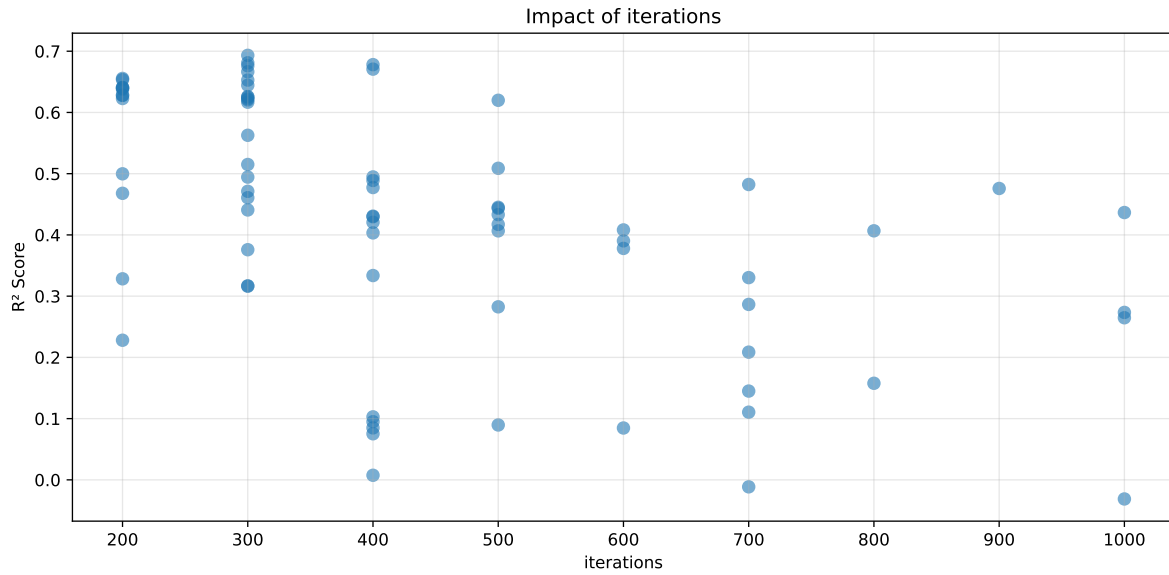
===== TOP Importance FEATURES =====

feature importance	r_umidity	21.821702	r_temperature_wl_12h_A_energy	11.273818
r_temperature_wl_2h_A_rms	9.787042	r_temperature	6.547728	r_CO2
5.276222	r_NH3_wl_12h_D1_energy	3.831322	r_umidity_wl_2h_A_energy_rel	3.655233
r_temperature_wl_2h_A_energy_rel	3.333037	r_NH3_wl_12h_D3_energy	3.251194	r_umid-
ity_wl_12h_A_rms	3.104017	r_temperature_wl_12h_A_rms	2.886408	r_NH3_wl_12h_D2_en-
ergy	2.385986	r_NH3	2.265835	r_NH3_wl_2h_A_energy
2.243232	r_NH3_wl_12h_A_en-	1.981635	r_NH3_wl_12h_D3_energy_rel	1.819326
r_THI	1.614755	r_umid-	ity_wl_2h_A_rms	1.387863
r_NH3_wl_2h_D1_energy	1.363357	r_NH3_wl_12h_A_rms	1.343627	r_umidity_wl_2h_A_energy
1.301922	r_NH3_wl_12h_A_energy_rel	1.209911	r_umidity_wl_12h_A_energy	1.080976
r_NH3_wl_2h_A_rms	1.018215	r_NH3_wl_2h_A_energy_rel	0.966267	r_temperature_wl_12h_D2_energy_rel
0.831928	r_umidity_wl_12h_D3_energy	0.719889	r_temperature_wl_2h_A_energy	0.662922
r_umidity_wl_12h_D2_energy	0.540161	r_temperature_wl_12h_D3_energy	0.494470	



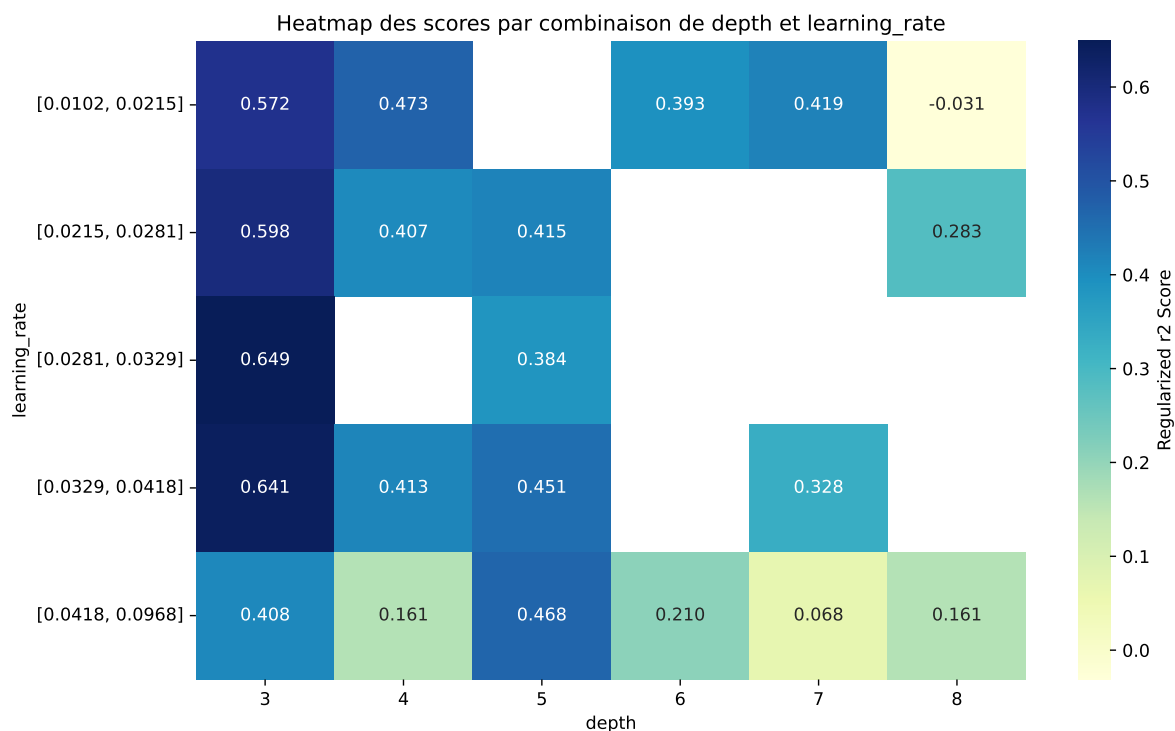
Paramètres avec >5% d'importance : ['depth', 'learning_rate', 'iterations', 'border_count']





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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.727	0.6932	0.7059	0.0361	0.6932	0.7137	0.0204	1.0295	1.0487
2	0.7117	0.6811	0.6903	0.0344	0.6811	0.6982	0.0172	1.0252	1.045
3	0.7079	0.6781	0.69	0.0357	0.6781	0.6963	0.0182	1.0268	1.044
4	0.7039	0.6762	0.6867	0.0348	0.6762	0.6949	0.0187	1.0277	1.041
5	0.6938	0.6707	0.6829	0.0334	0.6707	0.6886	0.0179	1.0267	1.0344
6	0.6905	0.6664	0.6772	0.0338	0.6664	0.6865	0.0201	1.0301	1.0361
7	0.6778	0.6555	0.6667	0.0361	0.6555	0.67	0.0145	1.0221	1.0339
8	0.6744	0.6536	0.6629	0.0342	0.6536	0.6708	0.0172	1.0264	1.032
9	0.6666	0.6529	0.6598	0.0326	0.6529	0.6645	0.0116	1.0178	1.0211
10	0.6609	0.6444	0.6522	0.0338	0.6444	0.6566	0.0123	1.019	1.0257
---	---	---	---	---	---	---	---	---	---
75	0.9522	0.7883	0.8111	0.0383	-0.0312	0.8302	0.0418	1.0531	1.2079

Hyperparameters:

Rank 1: {'reg__iterations': 300, 'reg__depth': 5, 'reg__learning_rate': 0.023368291742335526, 'reg__l2_leaf_reg': 29.620309130514308, 'reg__bagging_temperature': 3.079092268642844, 'reg__random_strength': 1.3638513547711677, 'reg__subsample': 0.8582895382545828, 'reg__min_data_in_leaf': 17, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.7359485263012993, 'reg__border_count': 71, 'reg__leaf_estimation_method': 'Newton',

'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 1421.9253283591397}

Rank 2: {'reg__iterations': 300, 'reg__depth': 5, 'reg__learning_rate': 0.028916713876385322, 'reg__l2_leaf_reg': 28.909040799287787, 'reg__bagging_temperature': 4.335647209384109, 'reg__random_strength': 4.570164832238113, 'reg__subsample': 0.8687250346914269, 'reg__min_data_in_leaf': 17, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.7373315791639503, 'reg__border_count': 39, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 4590.782952726475}

Rank 3: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.02396380246213454, 'reg__l2_leaf_reg': 43.36748511820322, 'reg__bagging_temperature': 4.588496692254097, 'reg__random_strength': 1.037327511549258, 'reg__subsample': 0.8235036674951218, 'reg__min_data_in_leaf': 13, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.8433118255682538, 'reg__border_count': 49, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 2121.471157815631}

Rank 4: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.03353130537960771, 'reg__l2_leaf_reg': 29.388859447898678, 'reg__bagging_temperature': 2.0033281622989616, 'reg__random_strength': 4.6213264717812965, 'reg__subsample': 0.5212527957213837, 'reg__min_data_in_leaf': 16, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.5979320765255616, 'reg__border_count': 41, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 1506.4171970641748}

Rank 5: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.03125782829970489, 'reg__l2_leaf_reg': 9.812708479389881, 'reg__bagging_temperature': 4.587731571145085, 'reg__random_strength': 4.869020431593264, 'reg__subsample': 0.5687841124569438, 'reg__min_data_in_leaf': 25, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.7729066597910472, 'reg__border_count': 58, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 3439.6600124669853}

Rank 6: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.02935539264627665, 'reg__l2_leaf_reg': 28.627710837087122, 'reg__bagging_temperature': 4.2463307466289475, 'reg__random_strength': 1.5596578261207648, 'reg__subsample': 0.7276608729808579, 'reg__min_data_in_leaf': 17, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.7506509097909503, 'reg__border_count': 70, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 5281.582853430174}

Rank 7: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.03249849432217156, 'reg__l2_leaf_reg': 48.72557982856813, 'reg__bagging_temperature': 4.319206773467393, 'reg__random_strength': 0.6985226731810514, 'reg__subsample': 0.6684199495618363,

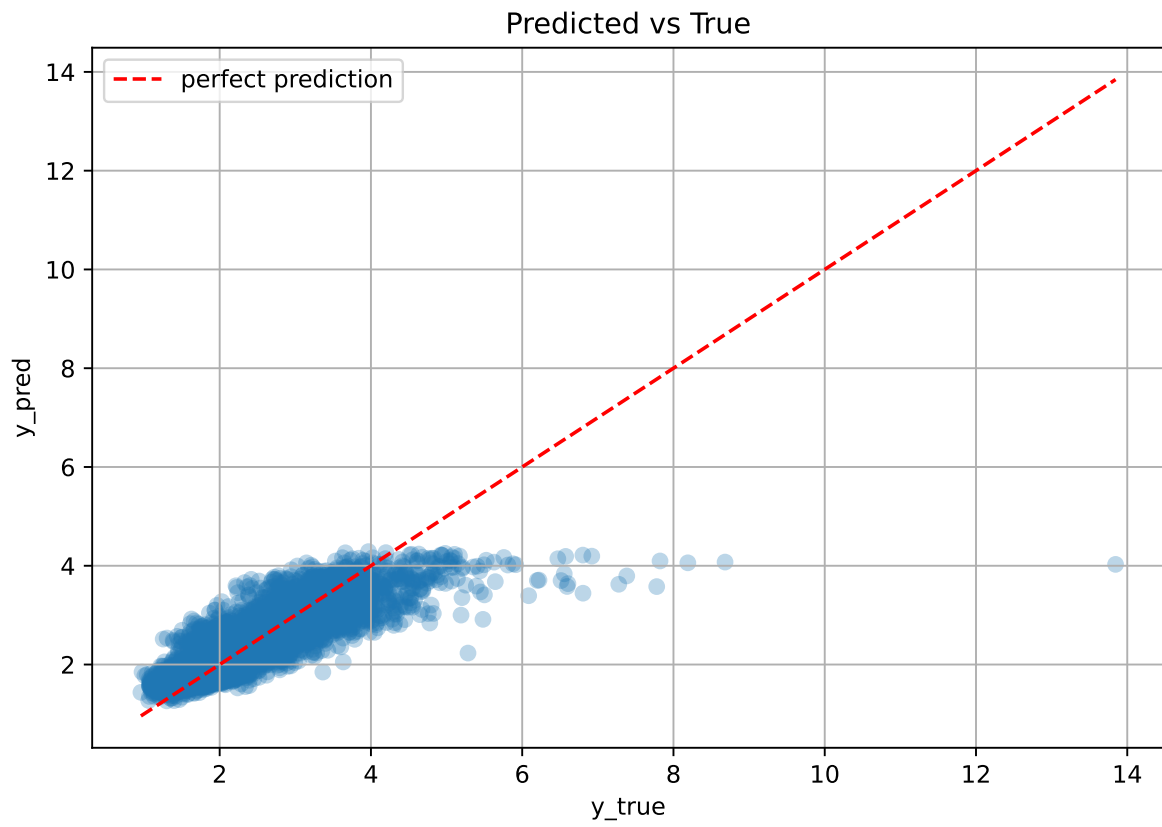
'reg__min_data_in_leaf': 14, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.8026214494668283, 'reg__border_count': 52, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 2177.810607884545}

Rank 8: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.032833338872561464, 'reg__l2_leaf_reg': 34.98422367081936, 'reg__bagging_temperature': 4.258023522916876, 'reg__random_strength': 0.7101537166681127, 'reg__subsample': 0.7945650968266753, 'reg__min_data_in_leaf': 15, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.8100893807773242, 'reg__border_count': 54, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 4067.8193497027924}

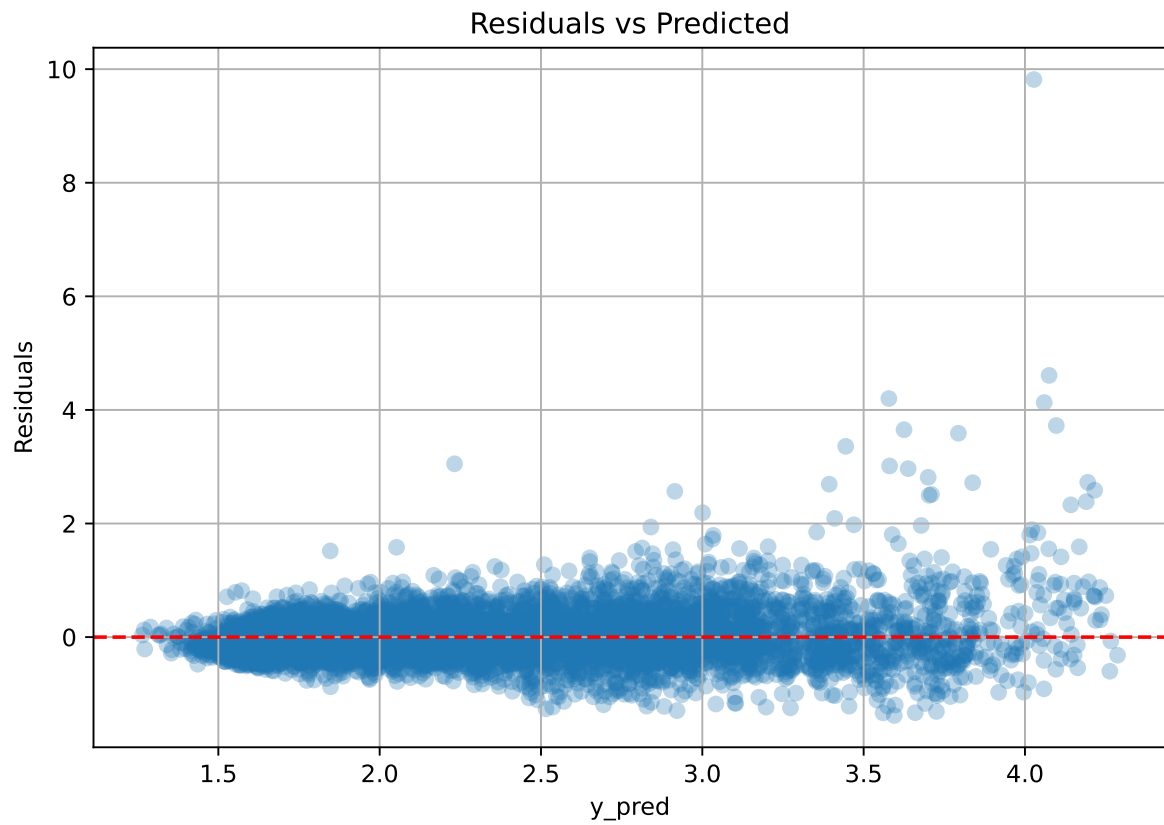
Rank 9: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.033104535990780865, 'reg__l2_leaf_reg': 40.21716020366938, 'reg__bagging_temperature': 4.176870608400011, 'reg__random_strength': 3.0468250398320036, 'reg__subsample': 0.7483480867947763, 'reg__min_data_in_leaf': 15, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.7682004571110751, 'reg__border_count': 53, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 5774.401586867263}

Rank 10: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.029568258073933873, 'reg__l2_leaf_reg': 1.0490279573625365, 'reg__bagging_temperature': 4.405070268159547, 'reg__random_strength': 4.591716674972548, 'reg__subsample': 0.7311639091928301, 'reg__min_data_in_leaf': 12, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.8641927497444271, 'reg__border_count': 39, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 5453.384588975537}

3.6.2 Scatter



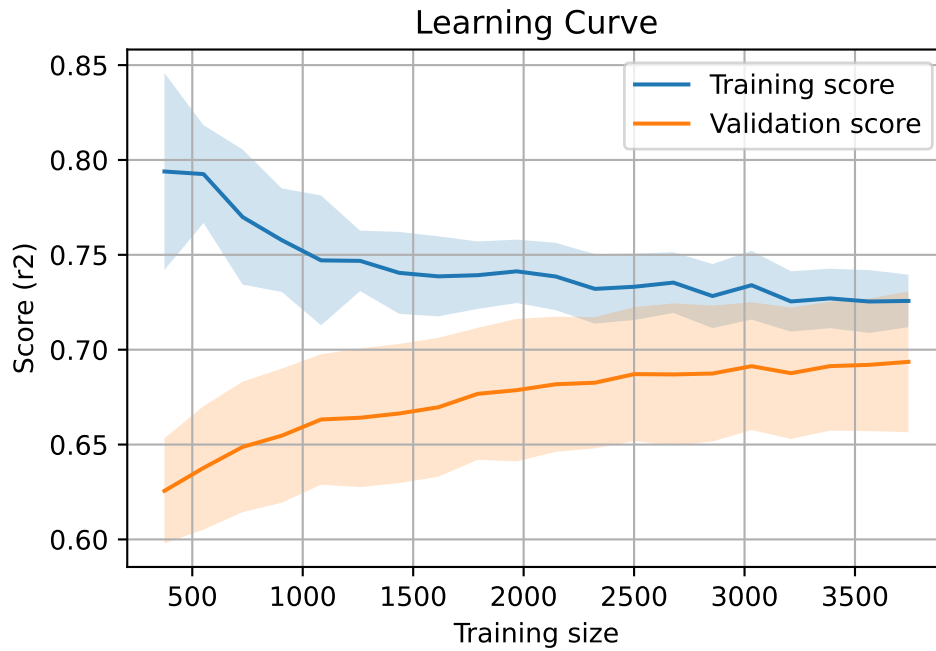
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 7639, `Len(training)` : 5348,

`Len(training_real)` : 3744, `Len(test_of_training)` : 1604, `Len(test)` : 2291



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.553	0.553	0.027	0.558	0.56	0.007	1.013	1.001	nan
LGBM	0.707	0.674	0.038	0.683	0.71	0.036	1.053	1.048	0.6745
CatBoost	0.727	0.693	0.036	0.706	0.714	0.02	1.03	1.049	0.6932

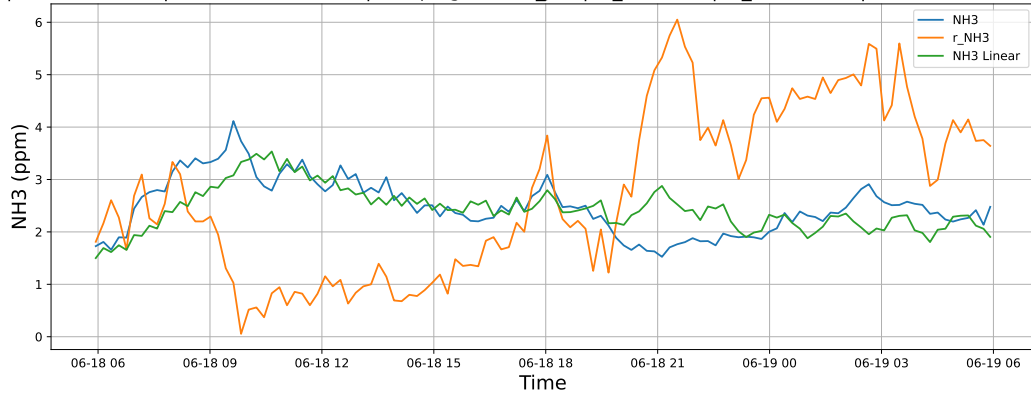
4 Plot

4.1 Standalone plots

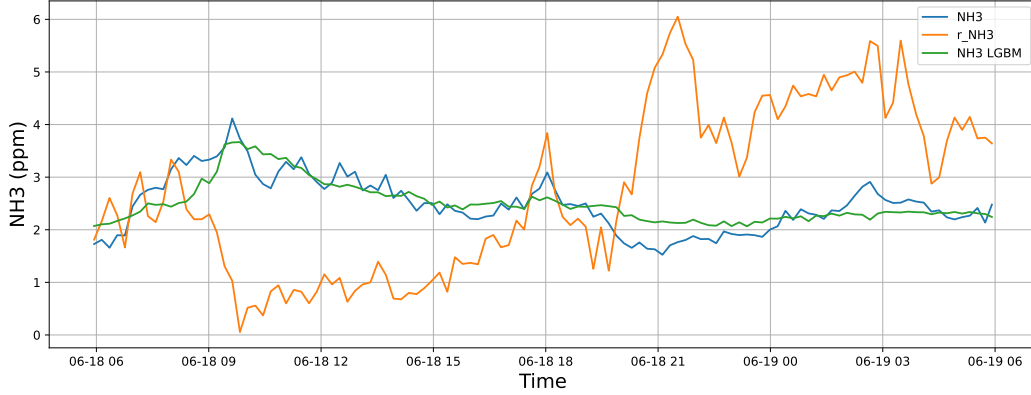
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

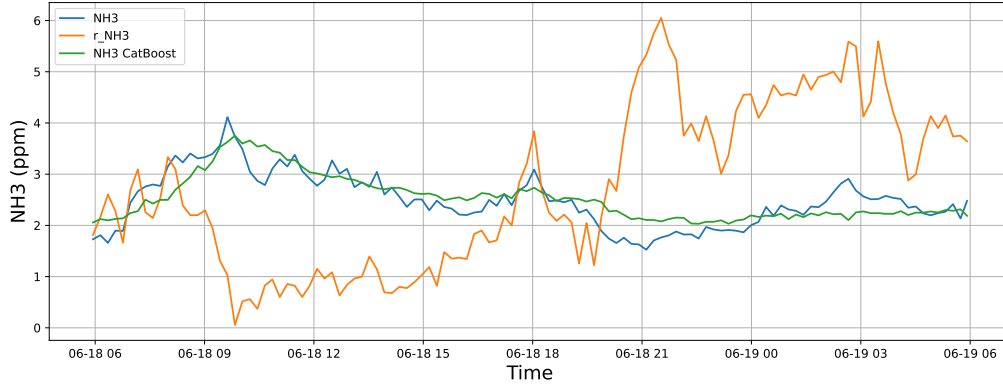
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

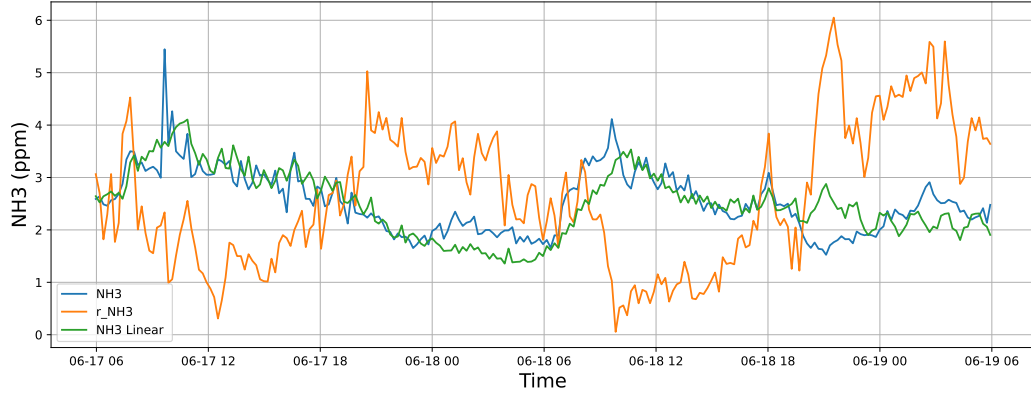


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

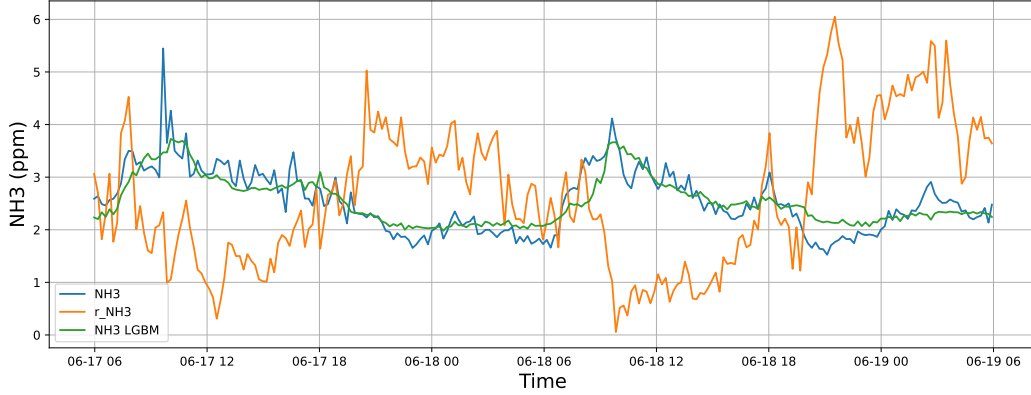


4.1.1.2 Time window : 48

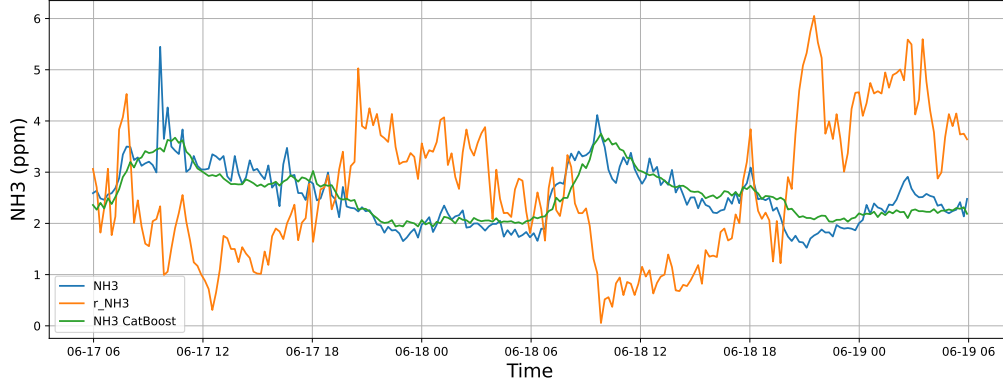
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

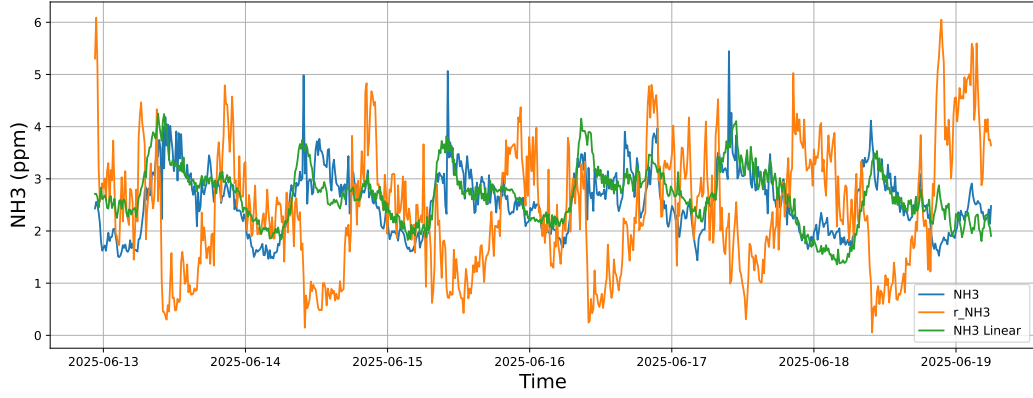


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

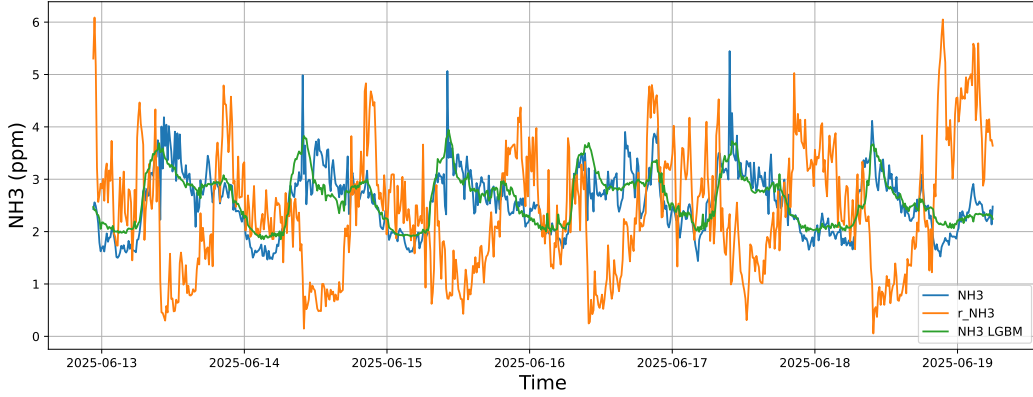


4.1.1.3 Time window : ALL

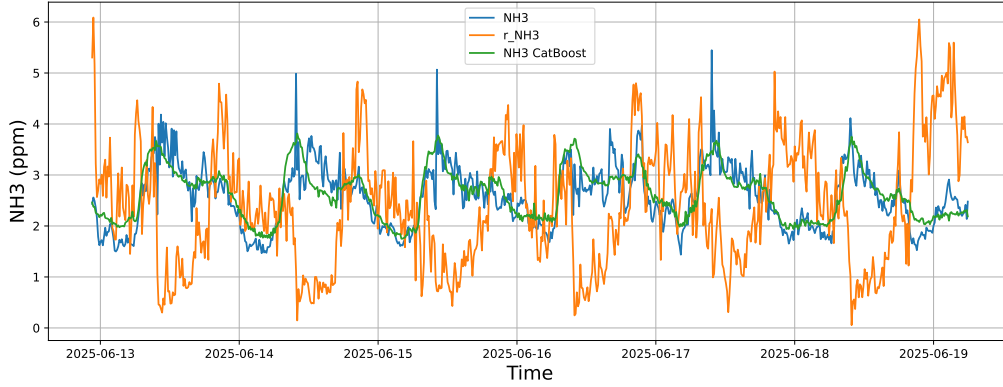
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



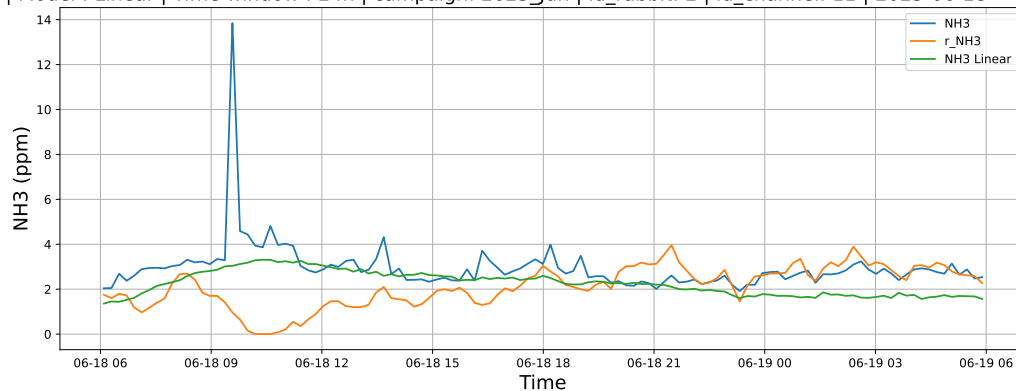
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



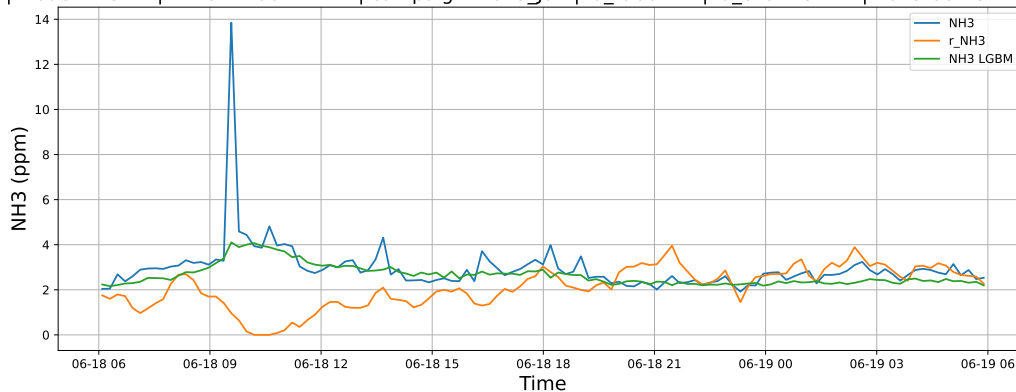
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

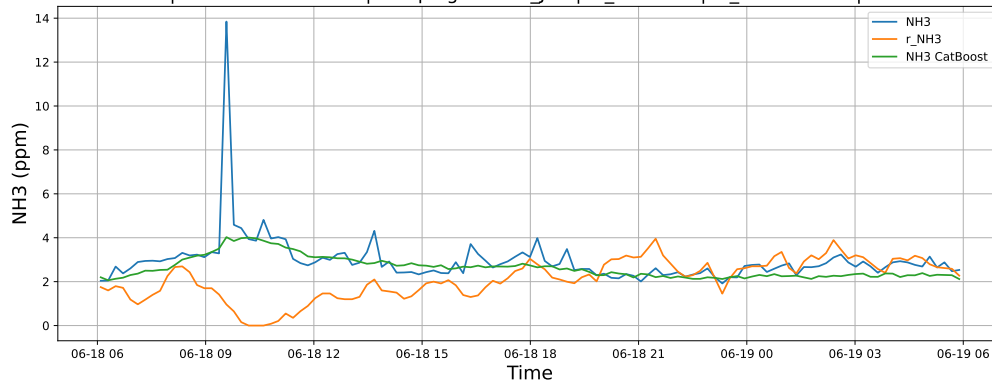
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

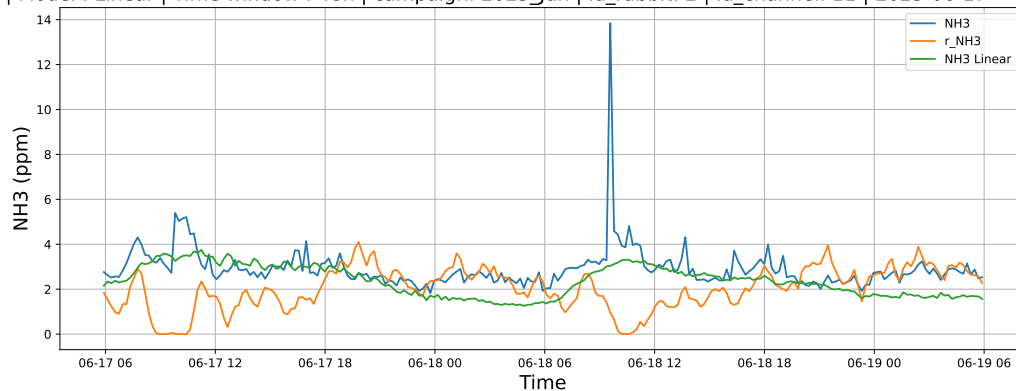


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

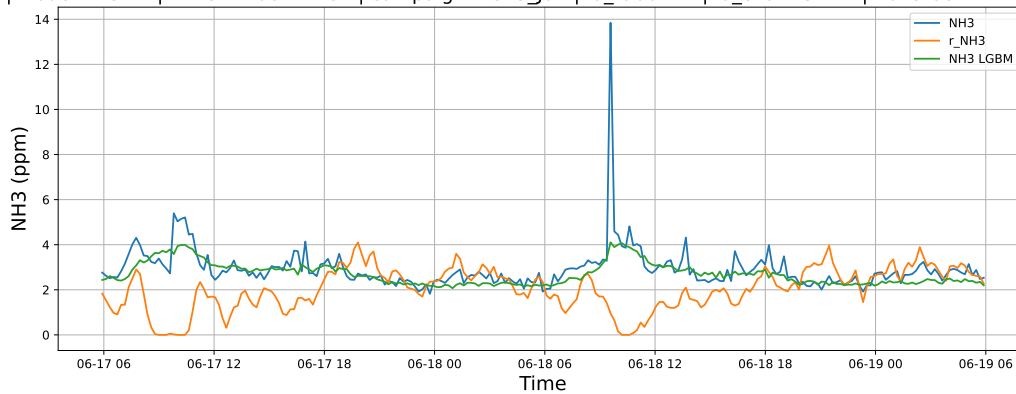


4.1.2.2 Time window : 48

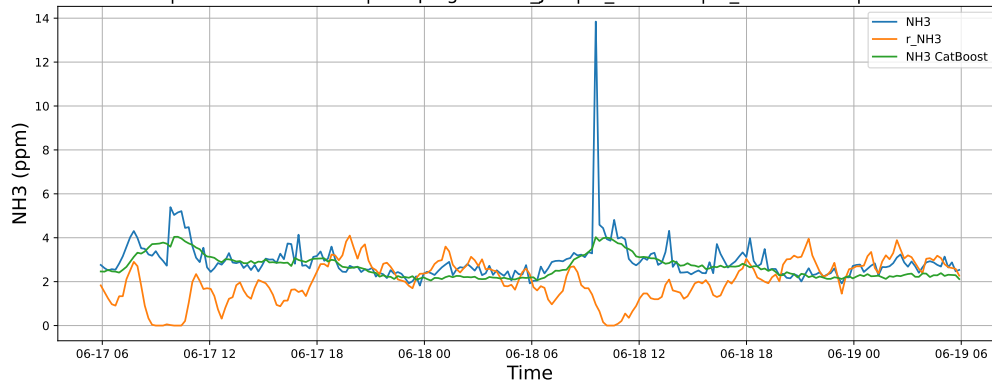
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

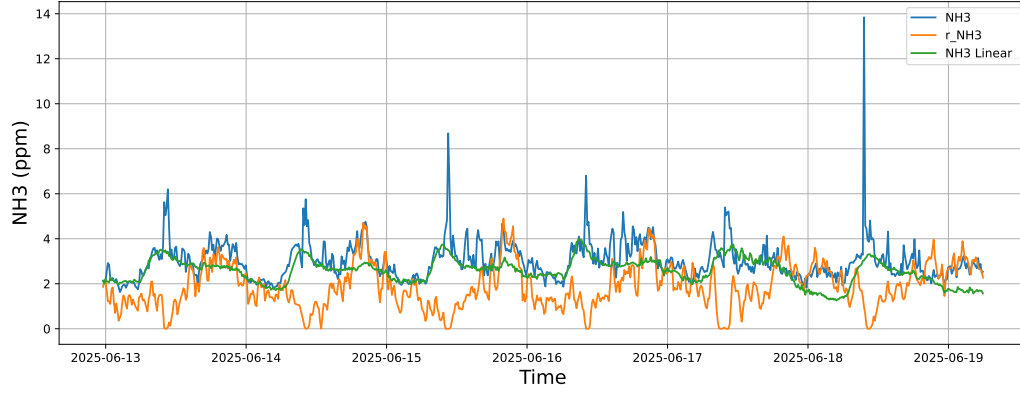


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

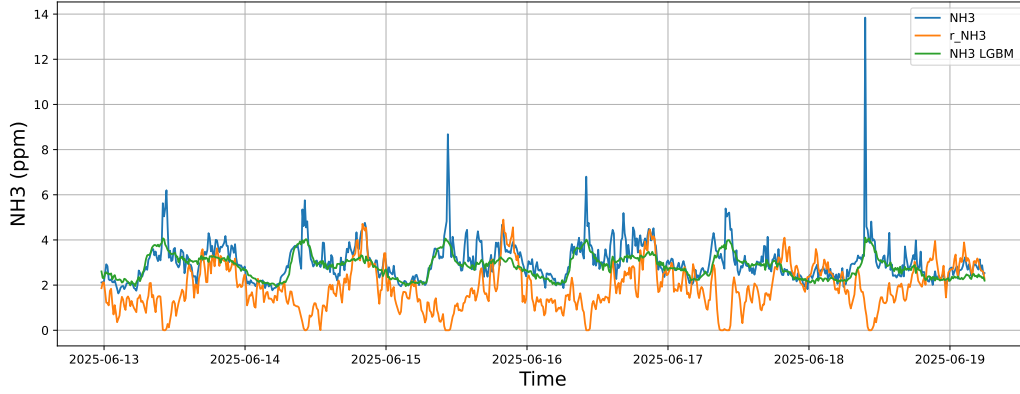


4.1.2.3 Time window : ALL

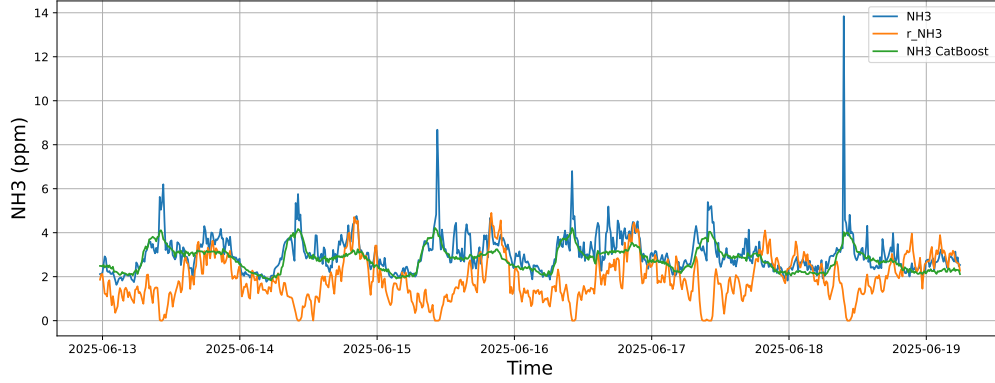
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



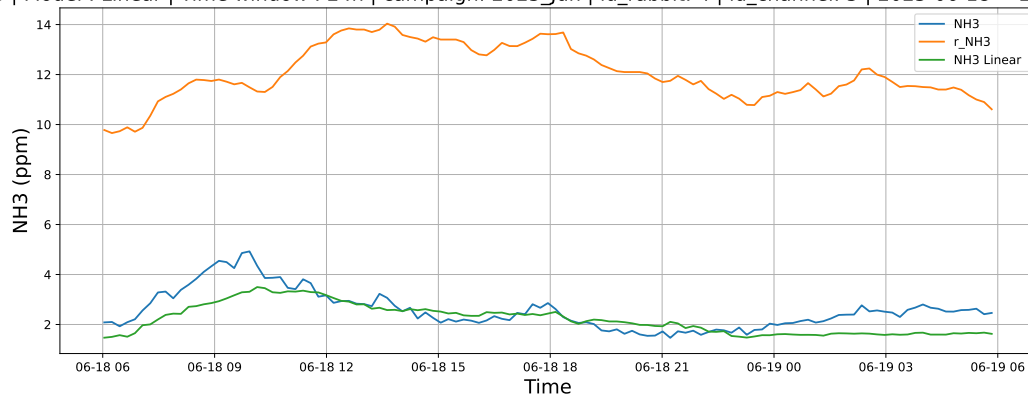
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



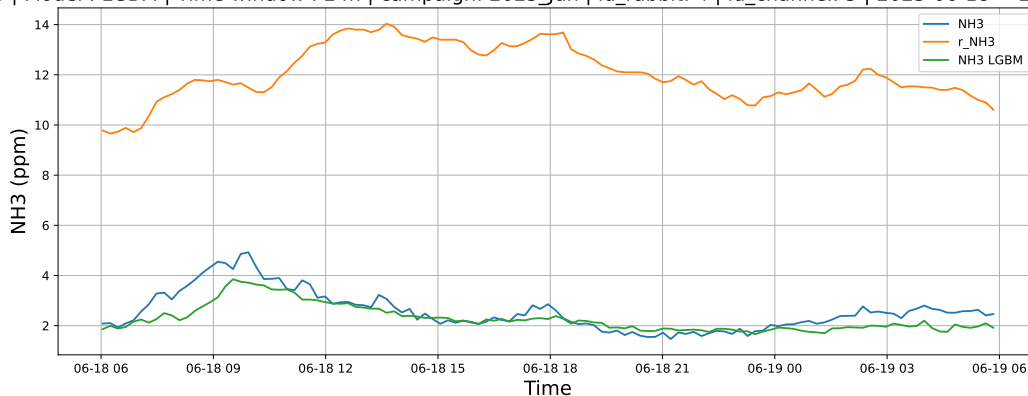
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

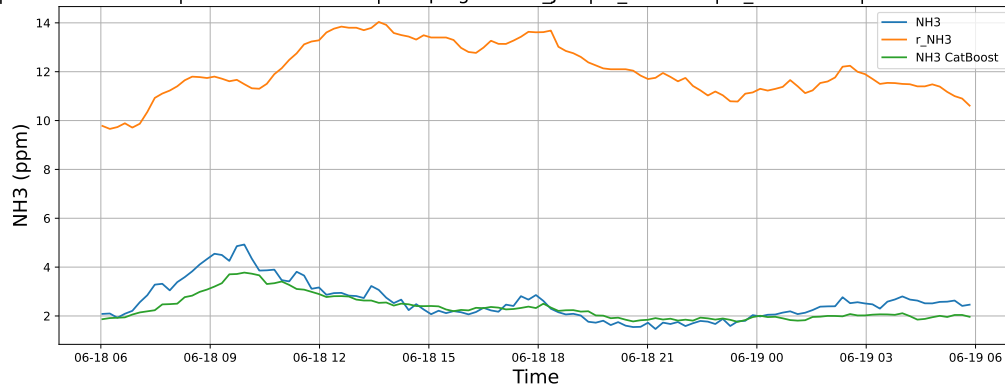
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

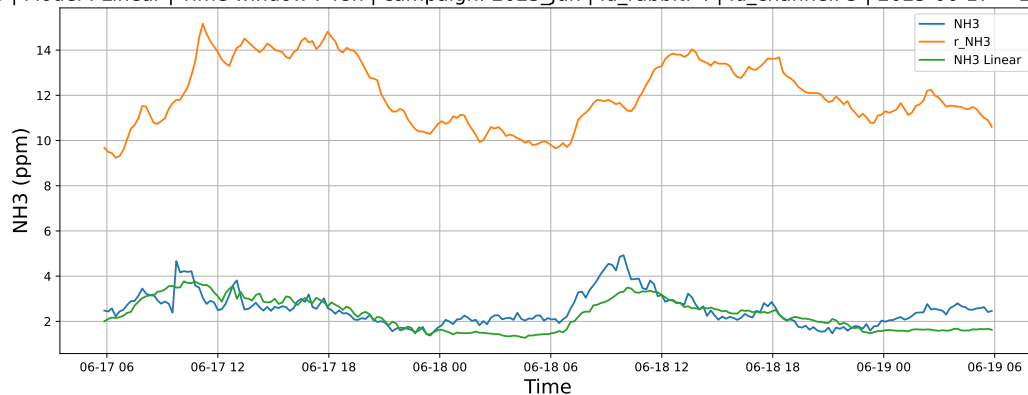


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

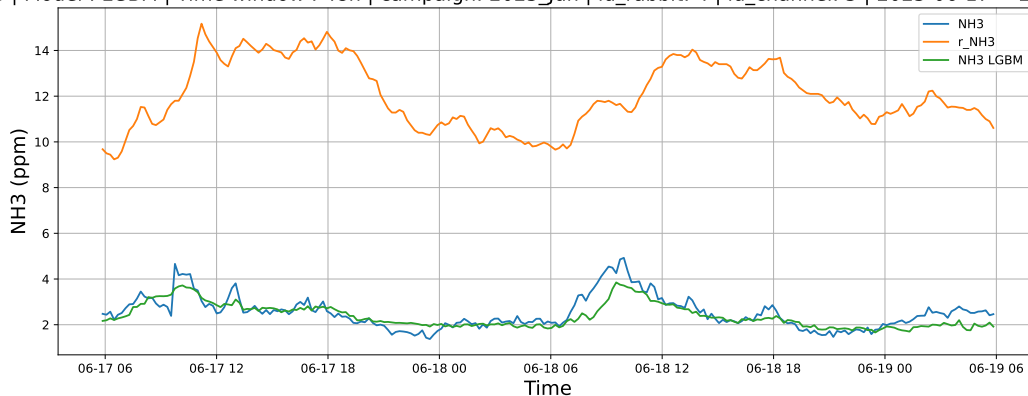


4.1.3.2 Time window : 48

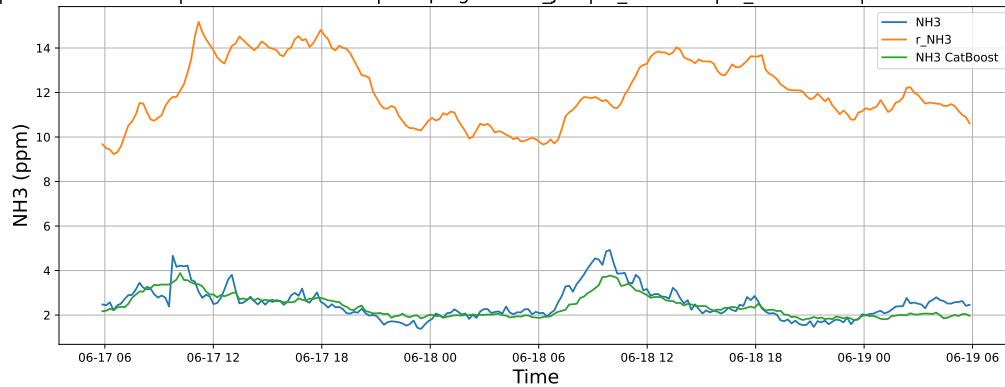
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

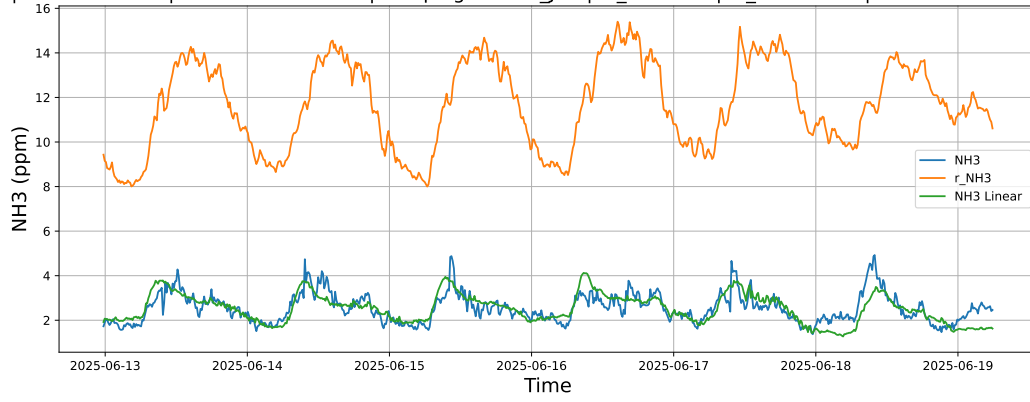


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

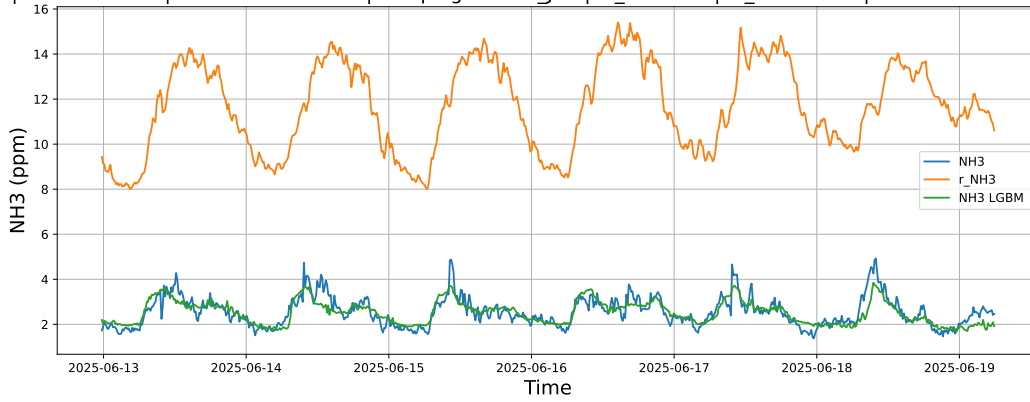


4.1.3.3 Time window : ALL

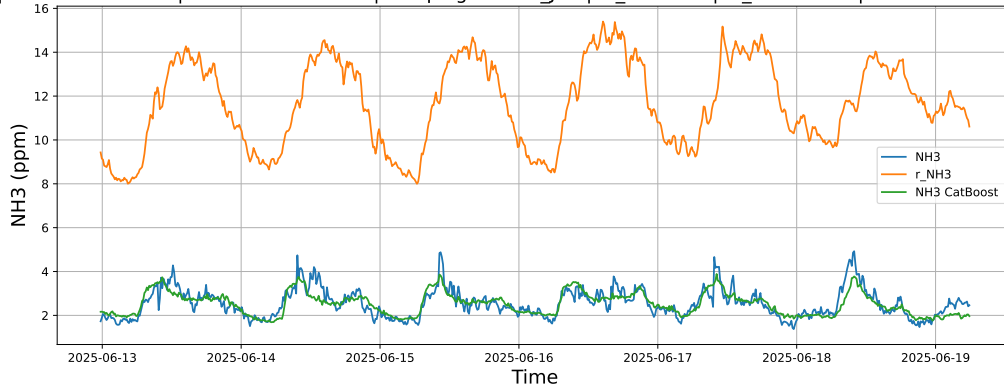
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



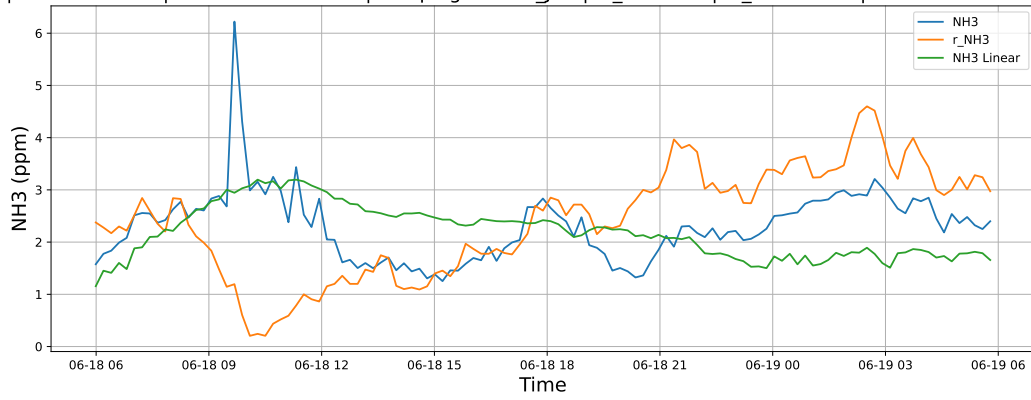
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



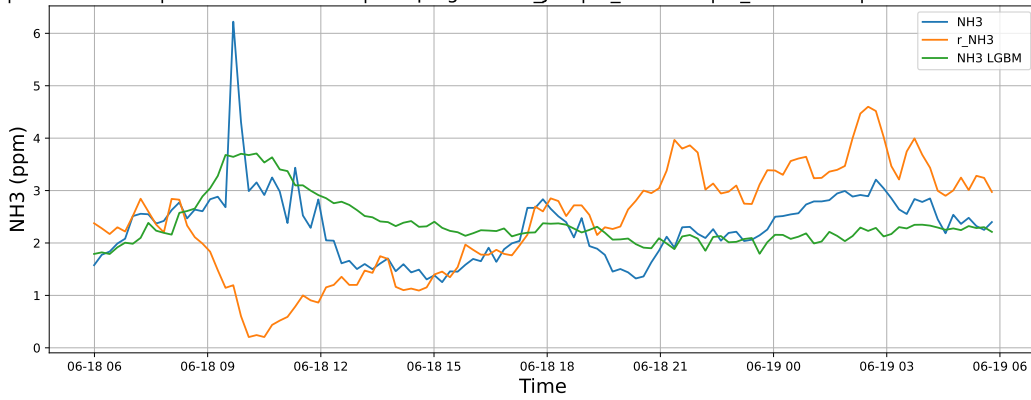
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

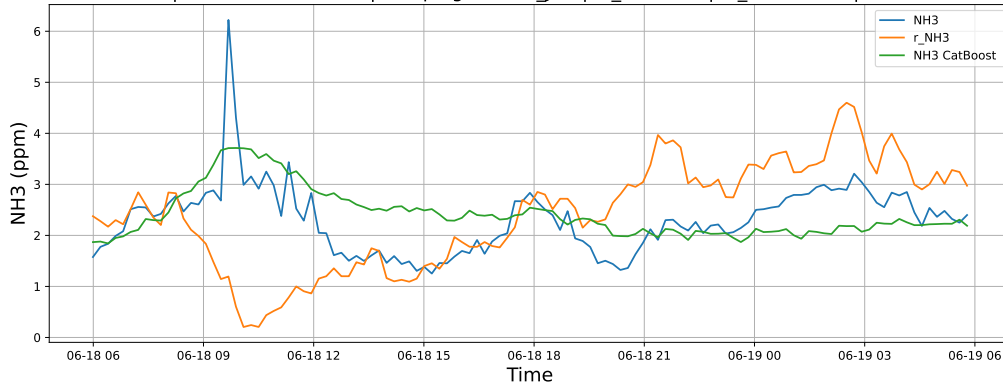
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

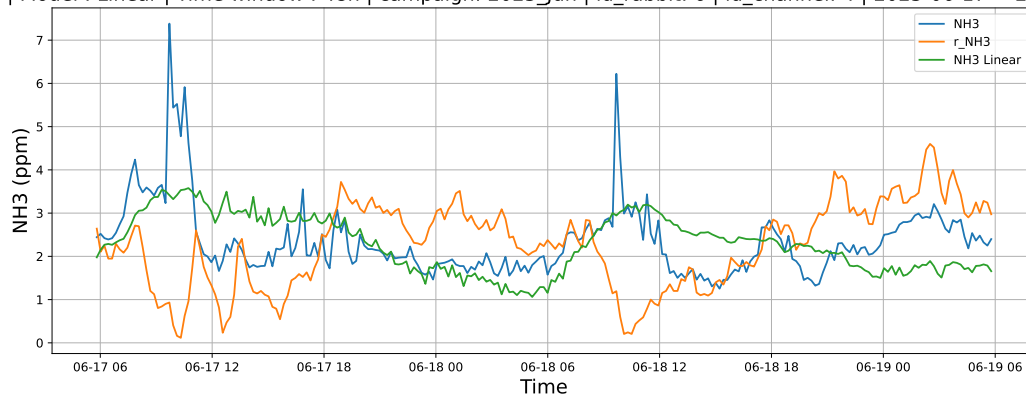


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

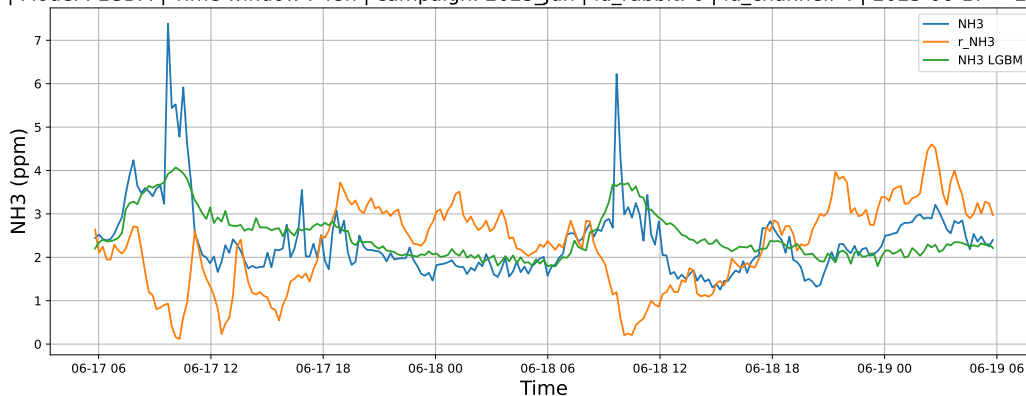


4.1.4.2 Time window : 48

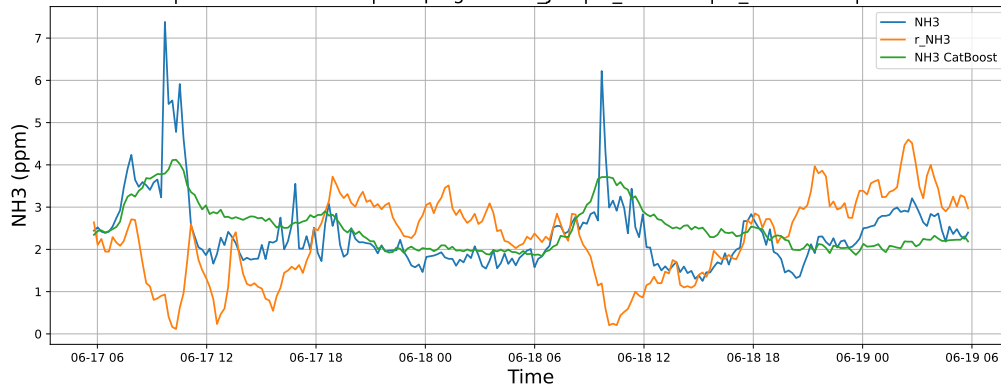
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

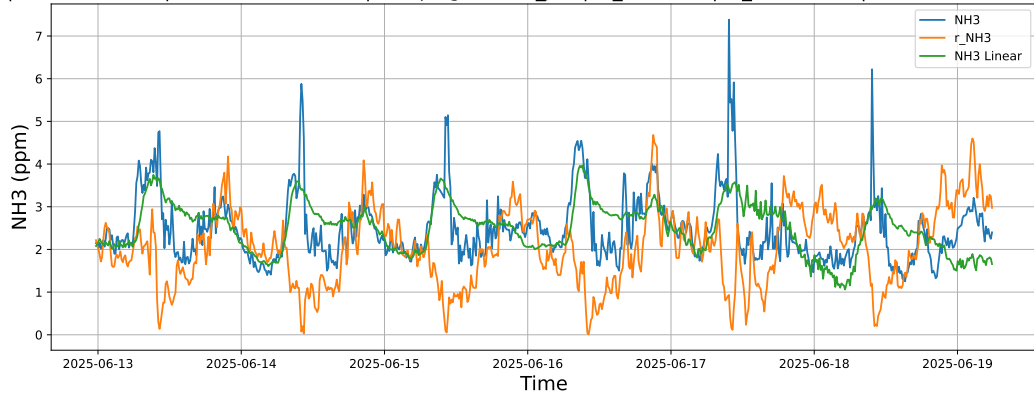


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

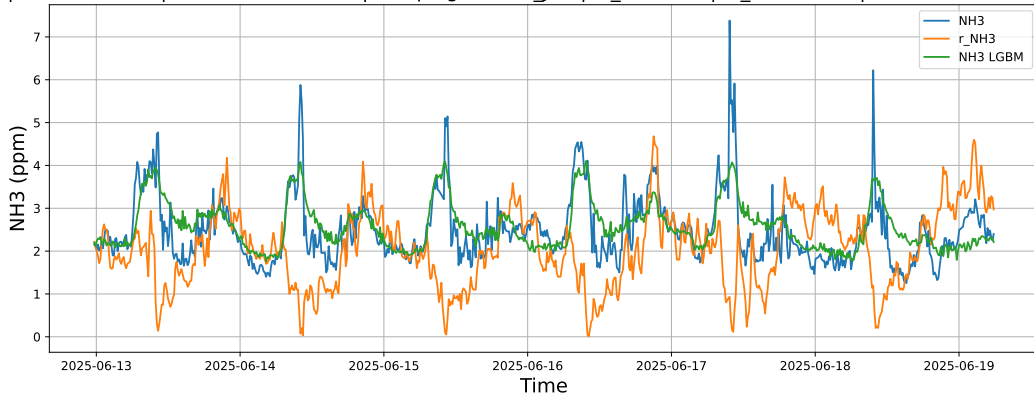


4.1.4.3 Time window : ALL

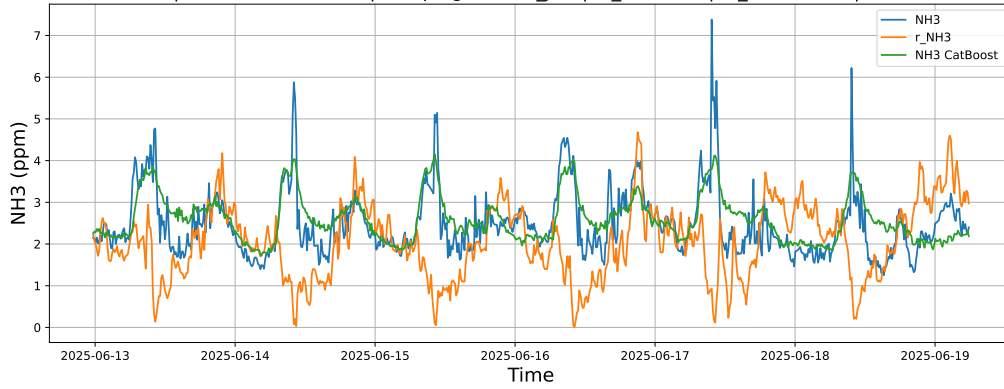
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



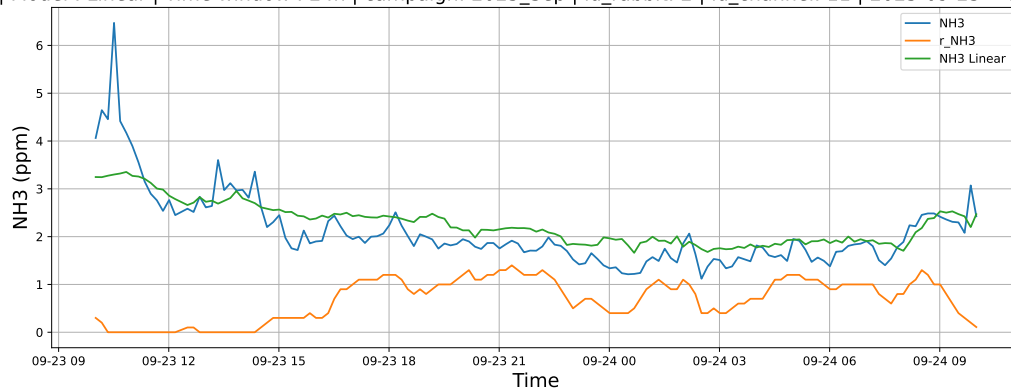
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



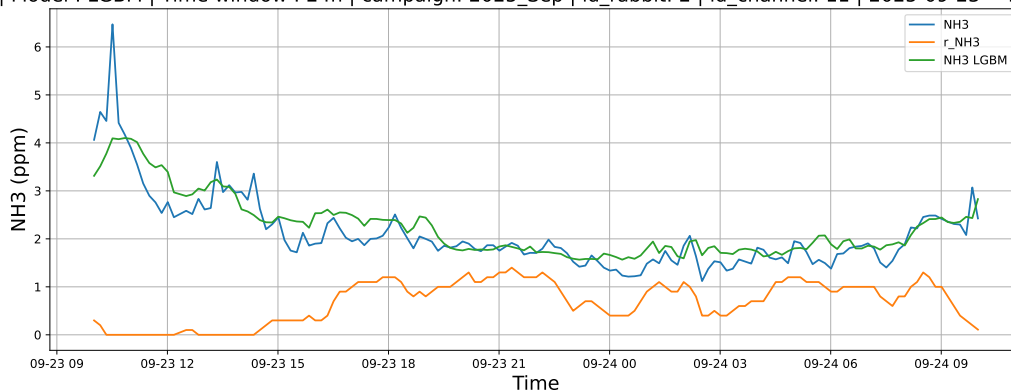
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

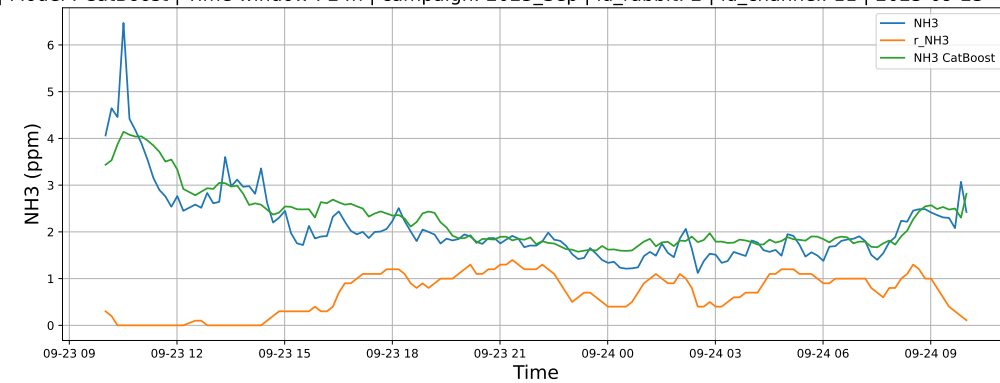
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

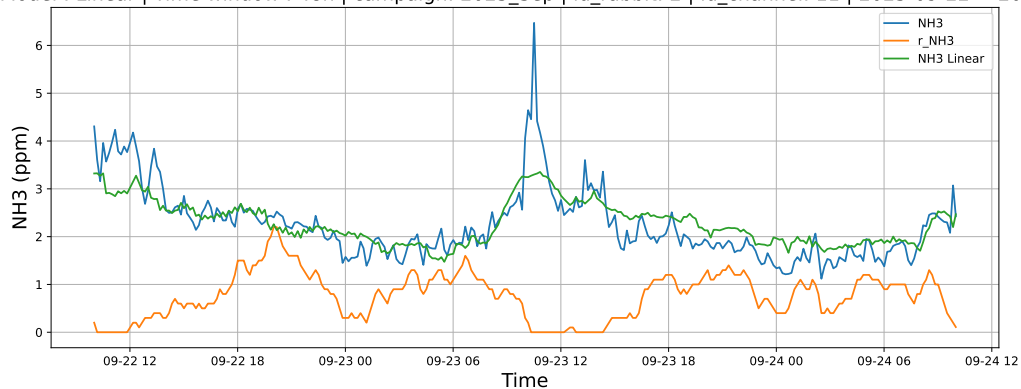


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

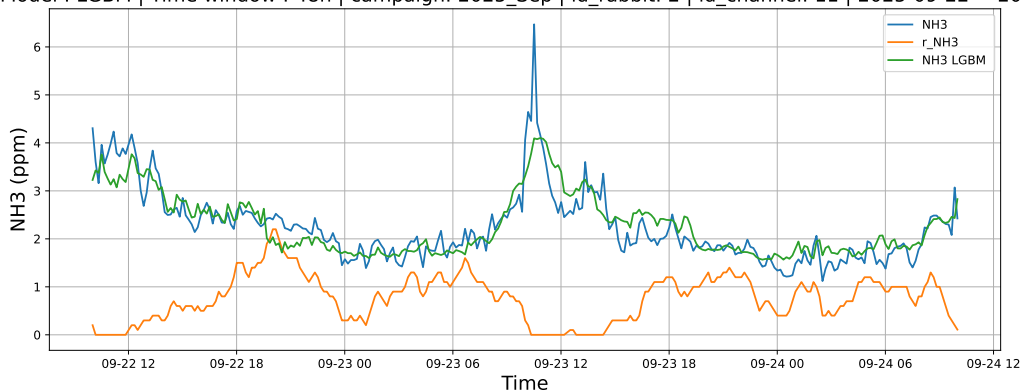


4.1.5.2 Time window : 48

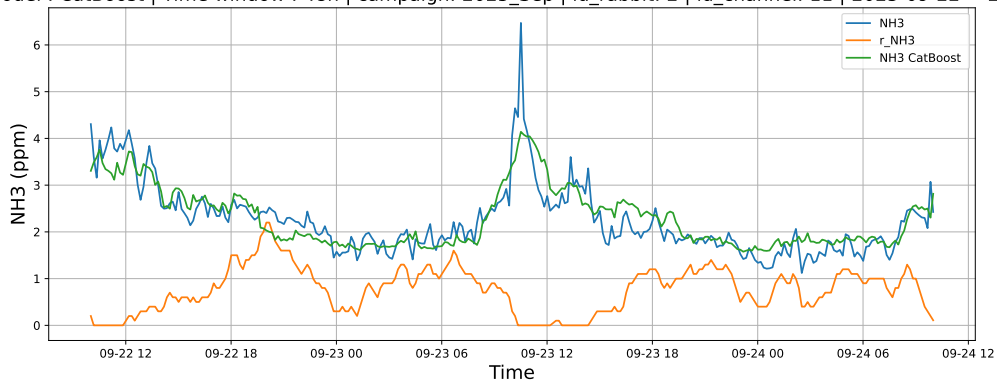
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

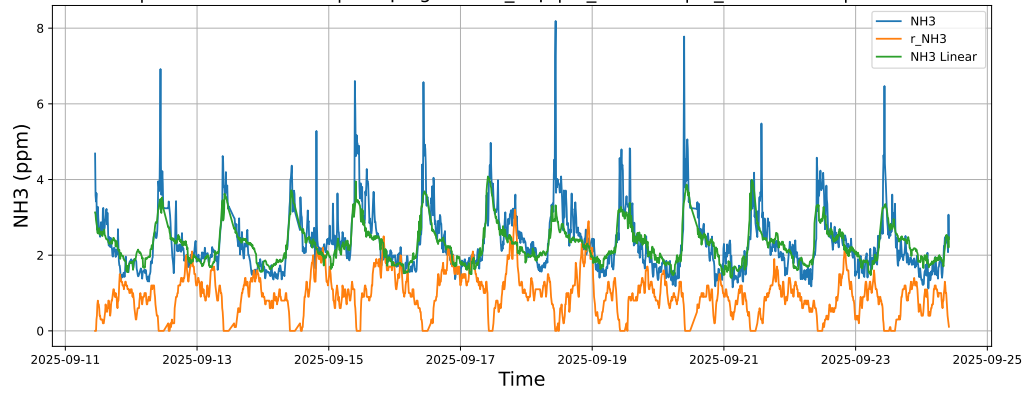


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

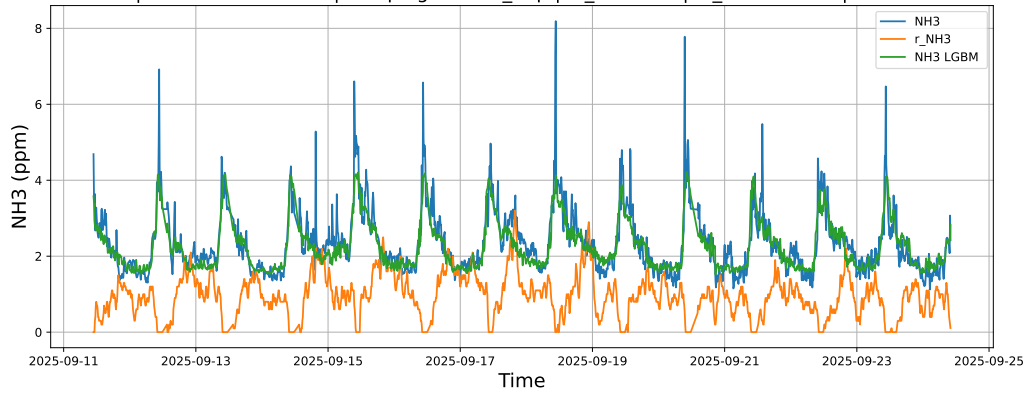


4.1.5.3 Time window : ALL

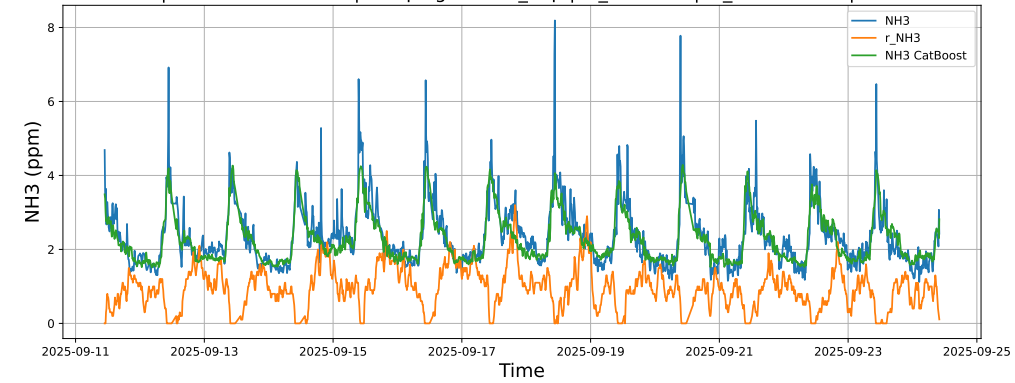
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



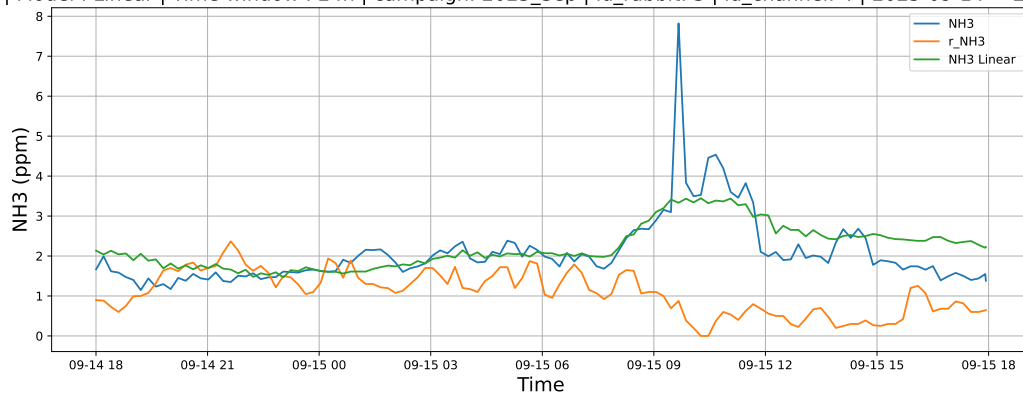
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



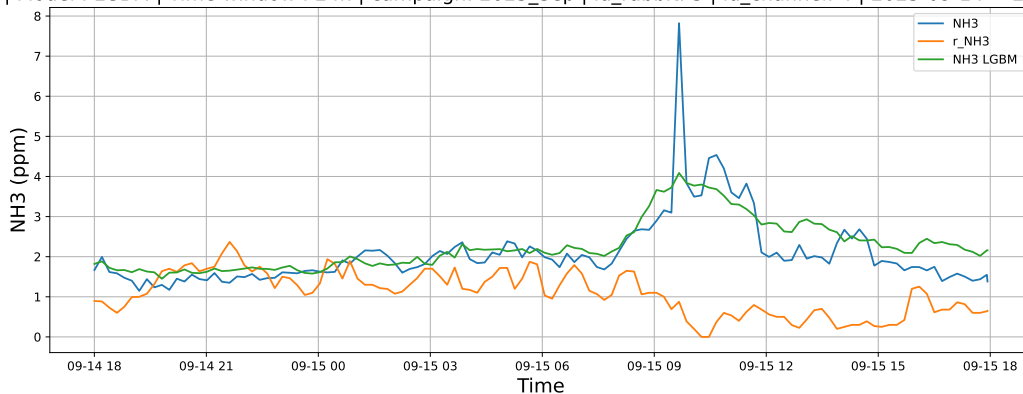
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

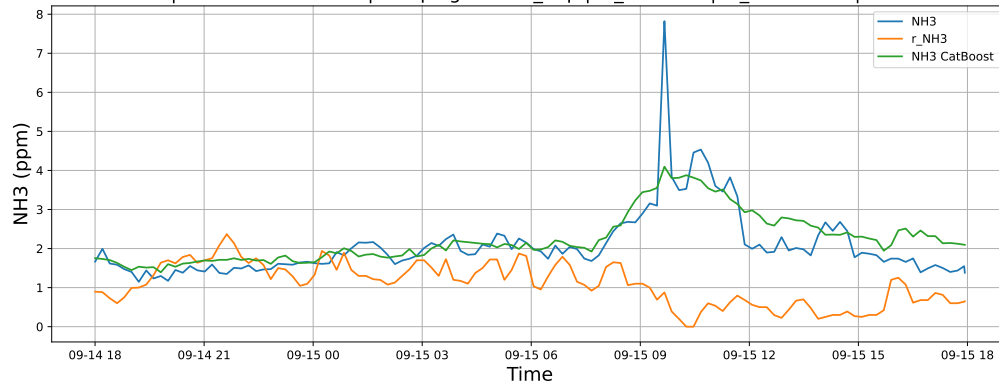
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

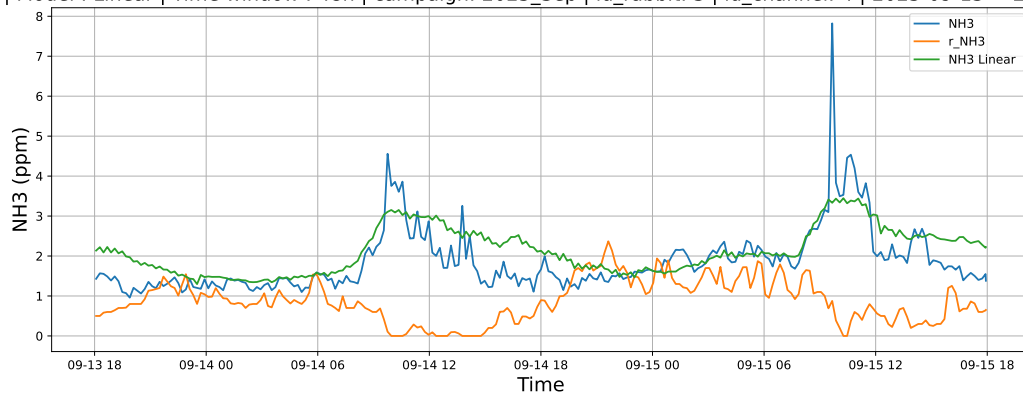


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

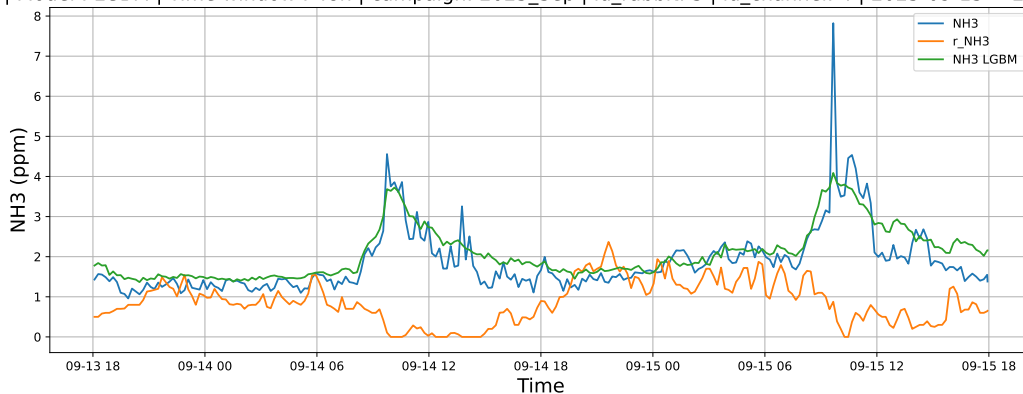


4.1.6.2 Time window : 48

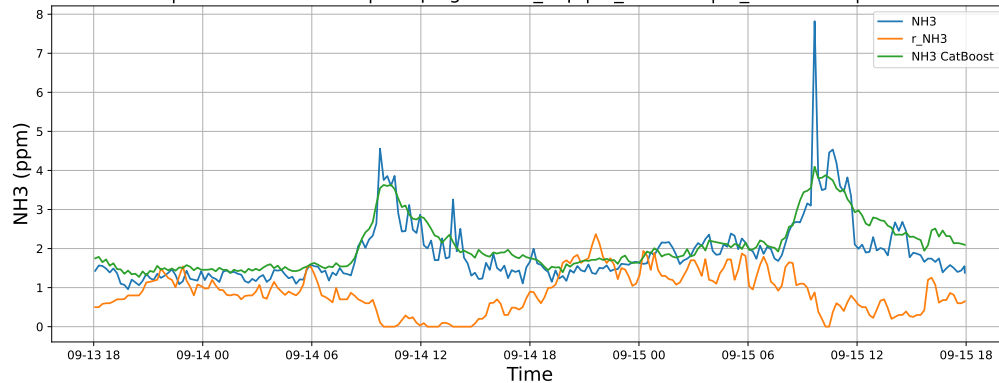
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

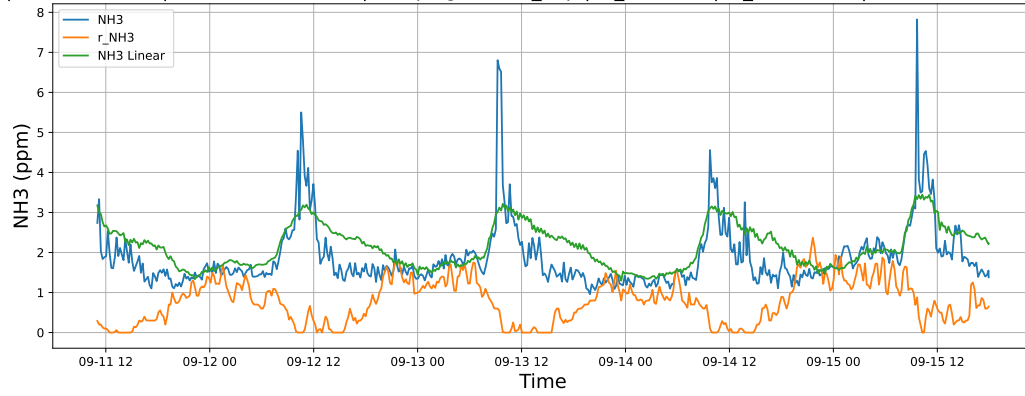


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

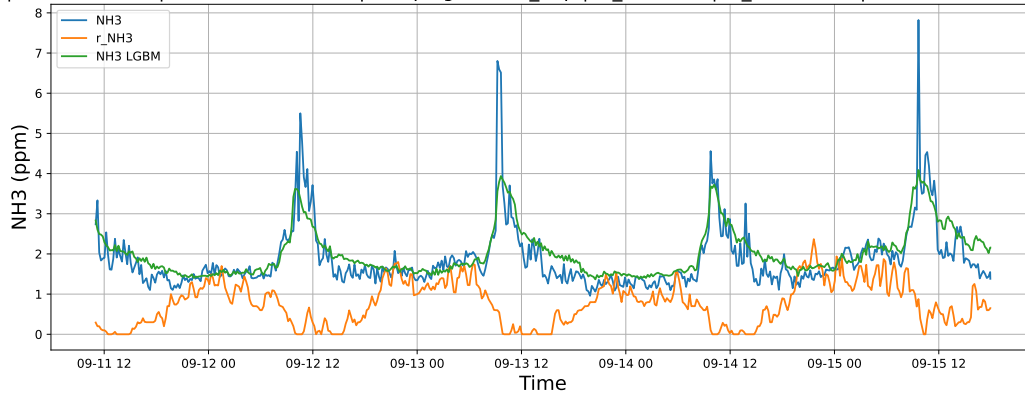


4.1.6.3 Time window : ALL

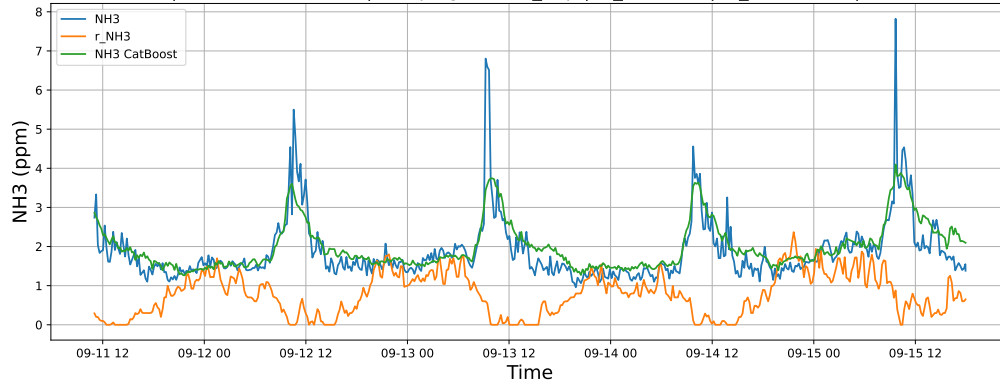
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



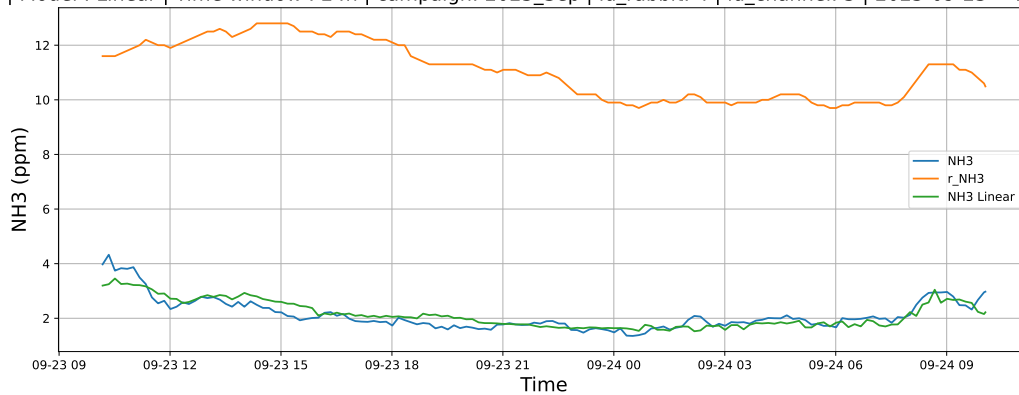
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



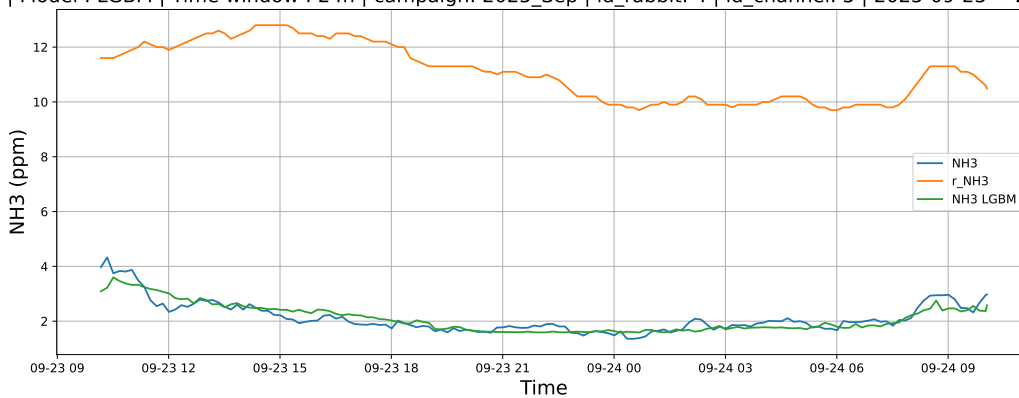
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

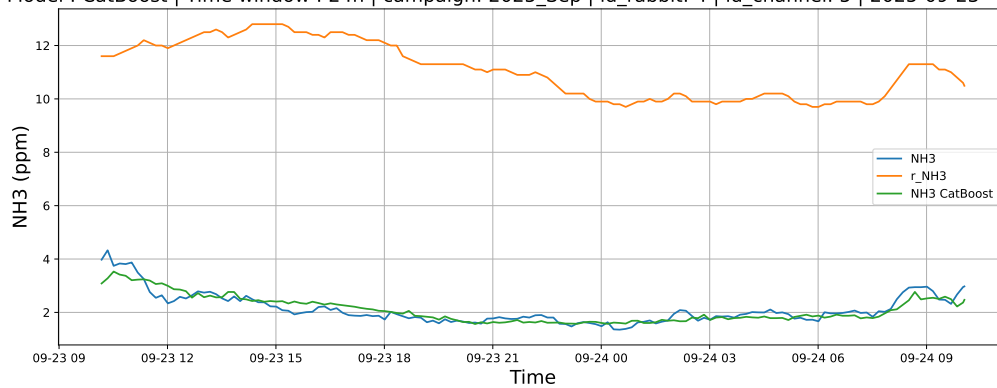
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

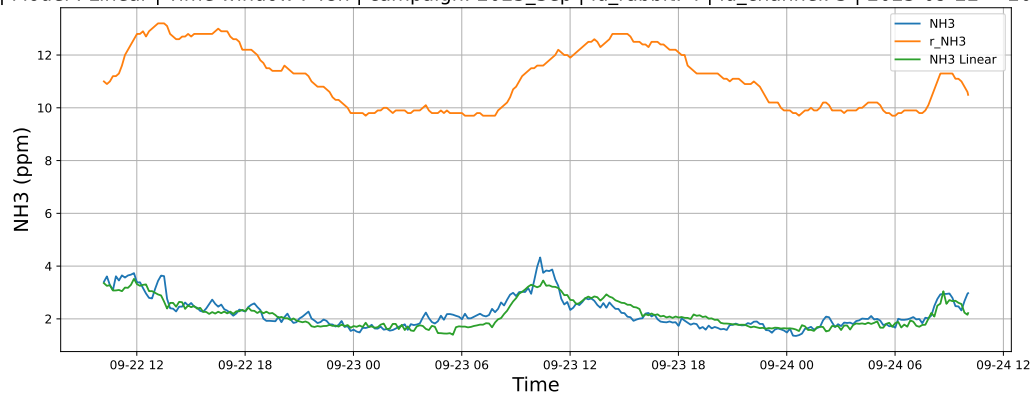


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

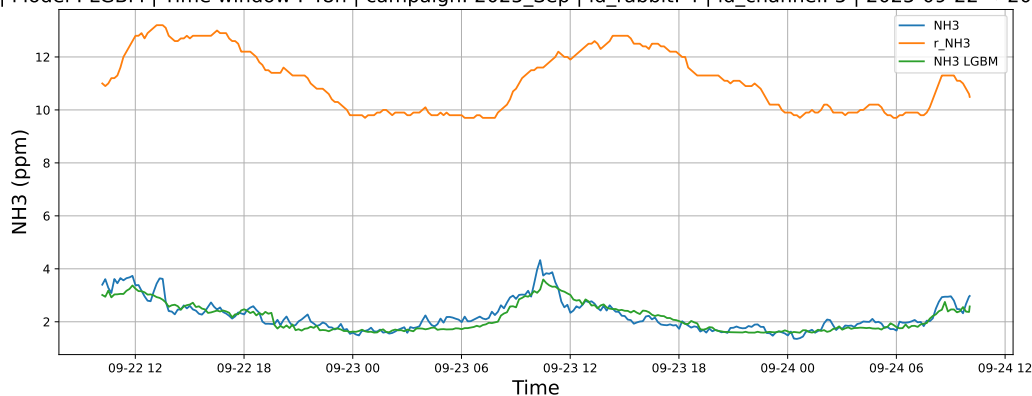


4.1.7.2 Time window : 48

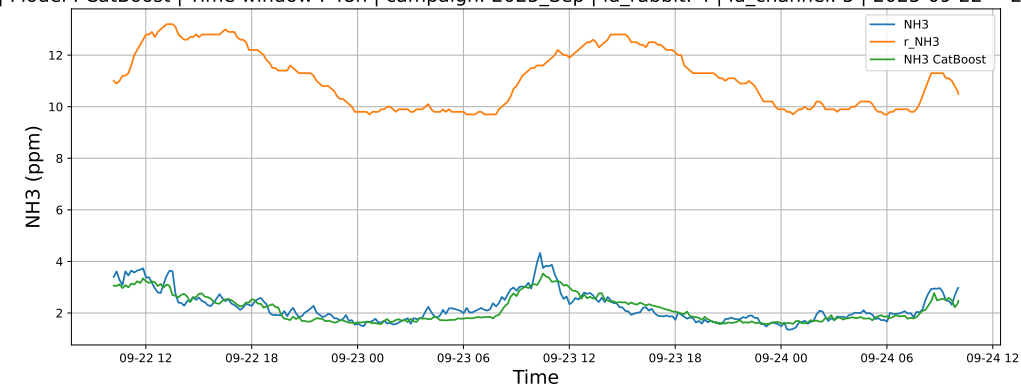
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

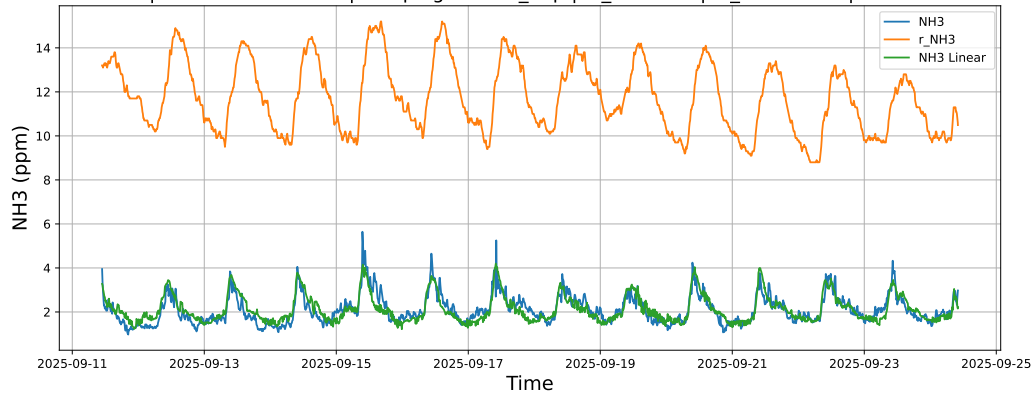


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

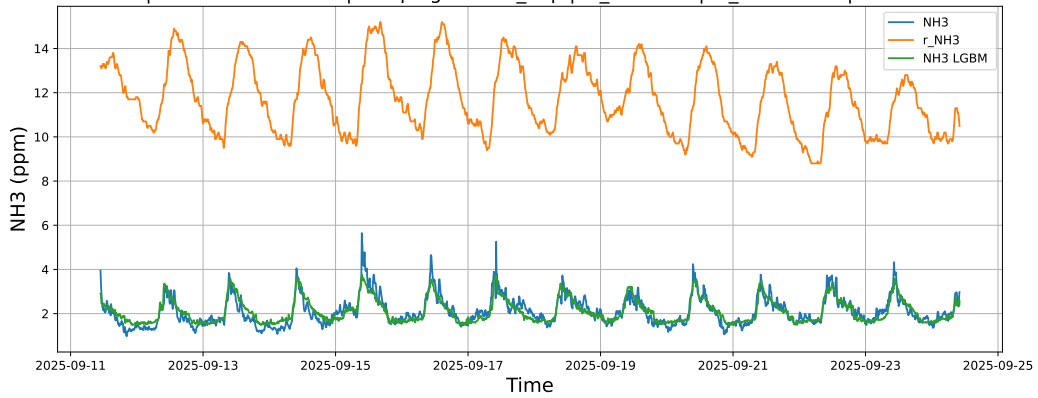


4.1.7.3 Time window : ALL

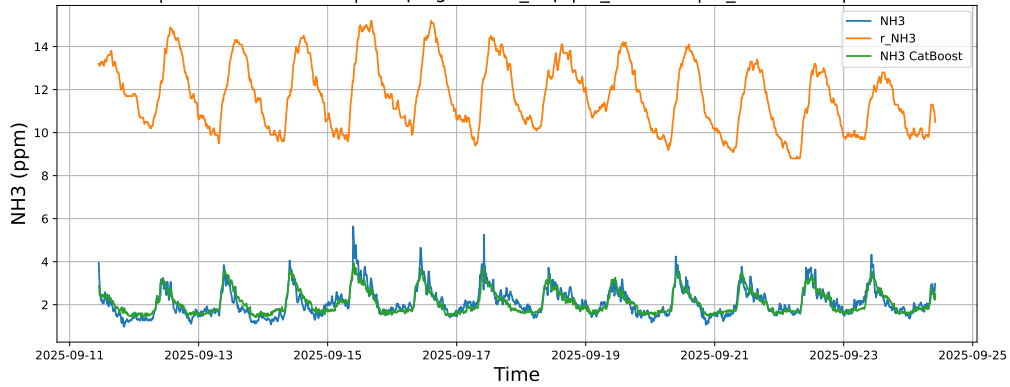
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



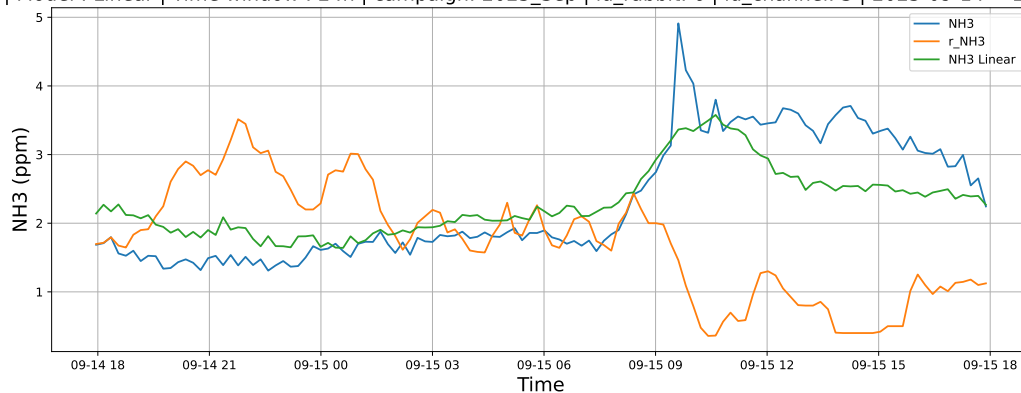
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



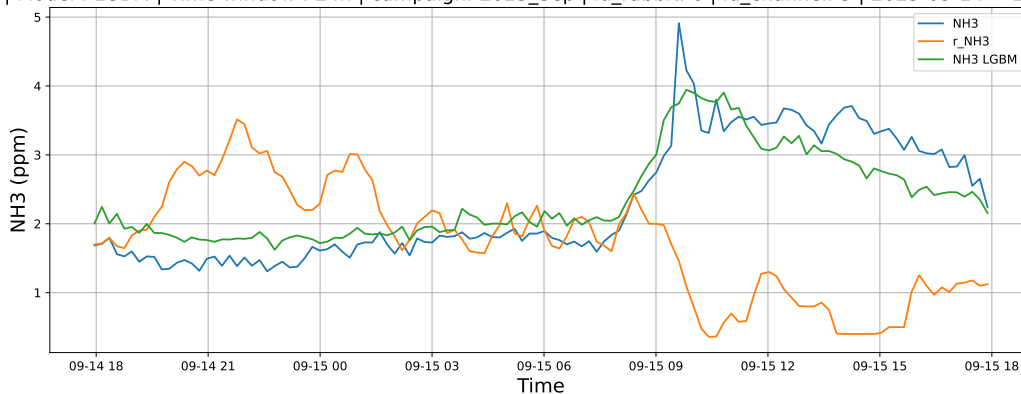
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

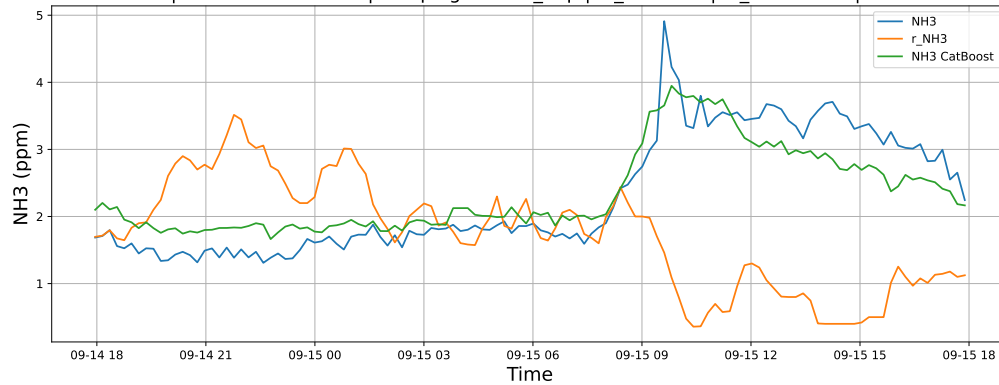
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

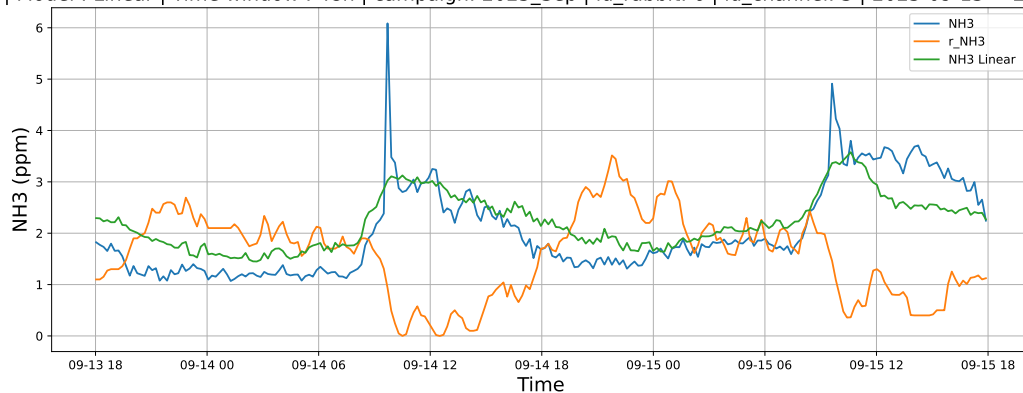


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

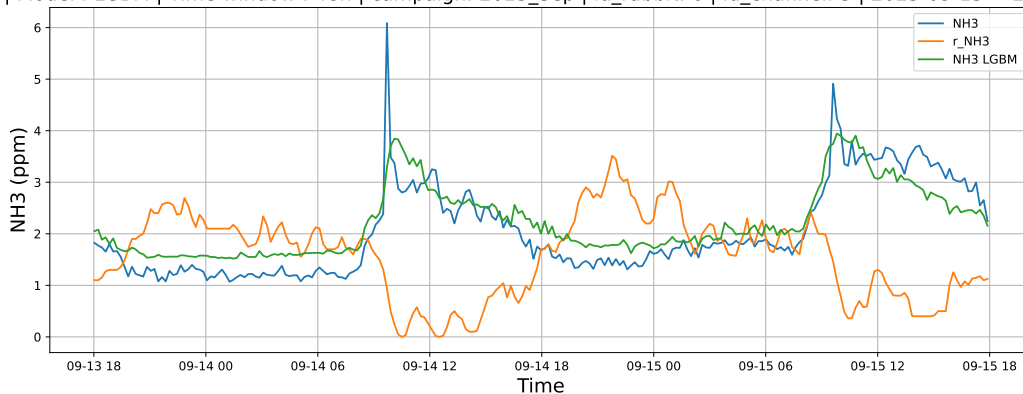


4.1.8.2 Time window : 48

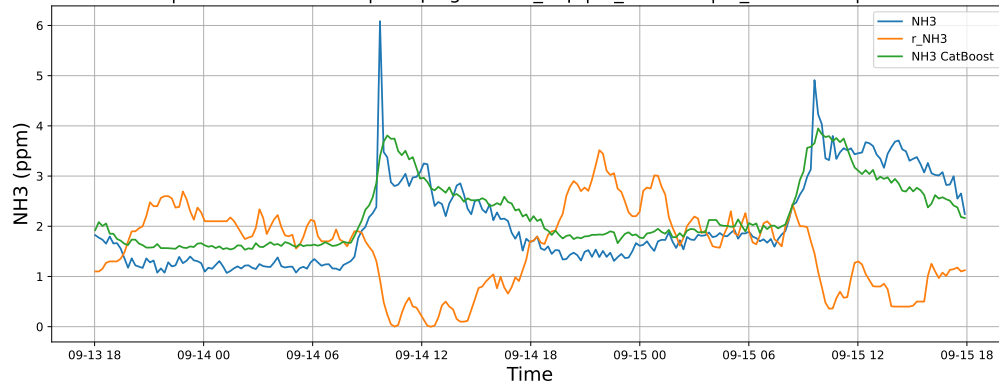
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

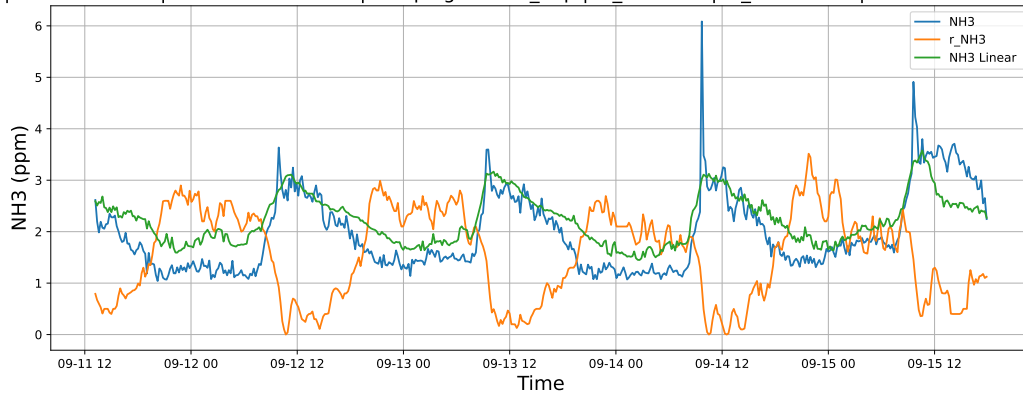


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

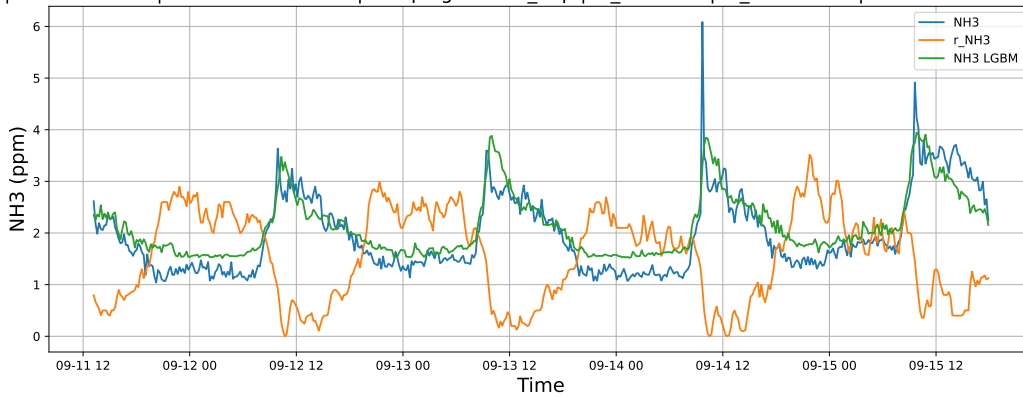


4.1.8.3 Time window : ALL

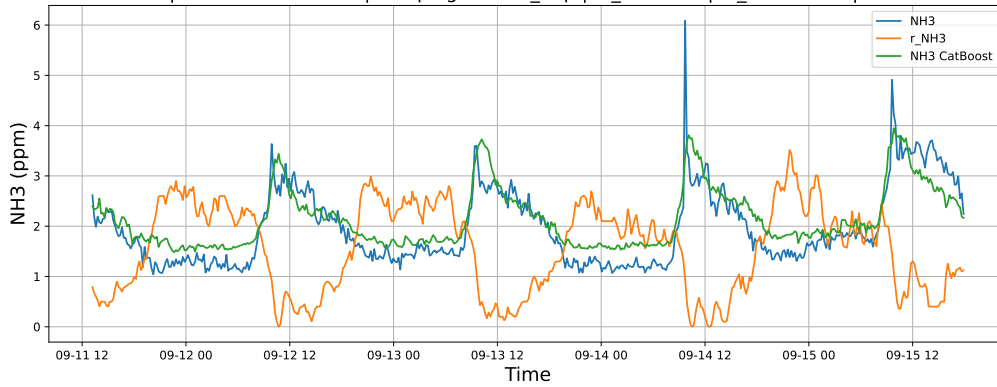
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

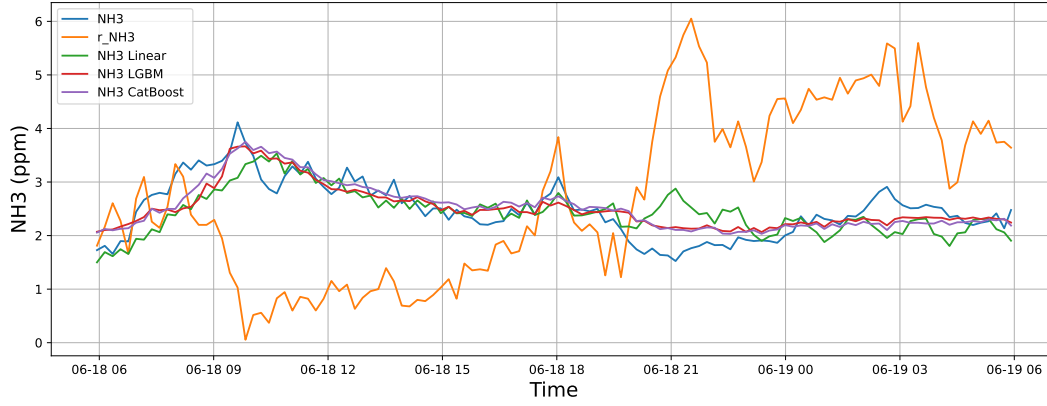


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

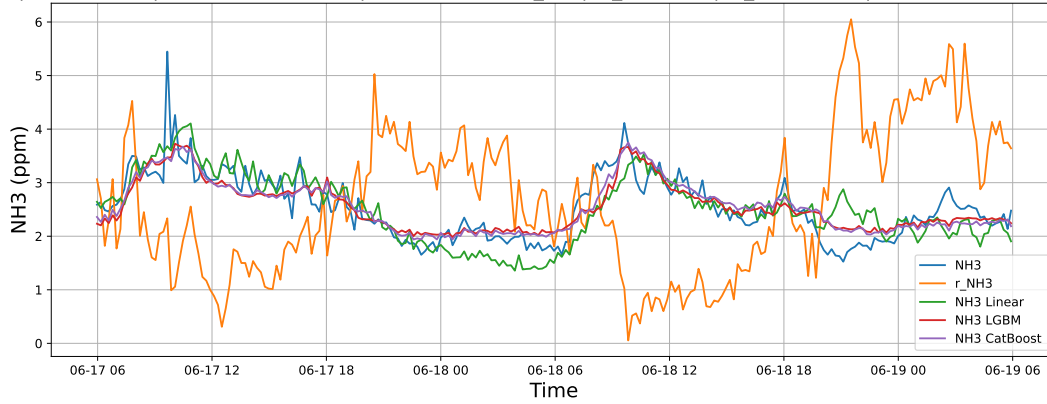
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



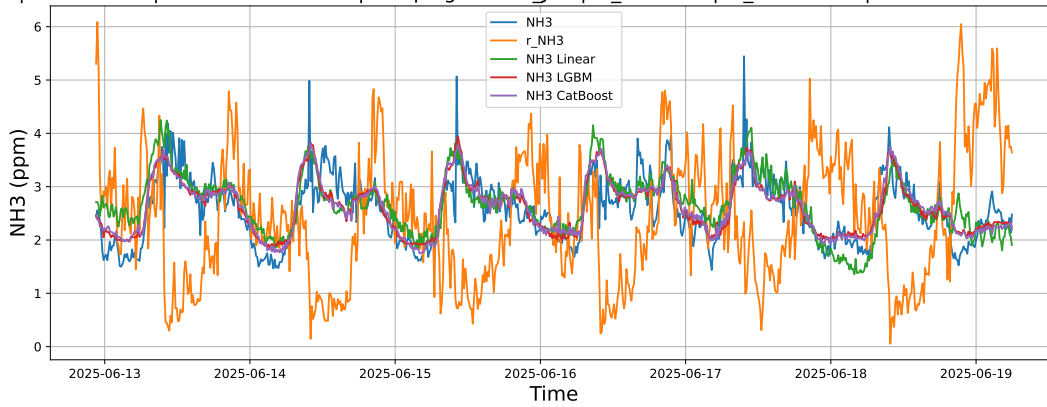
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

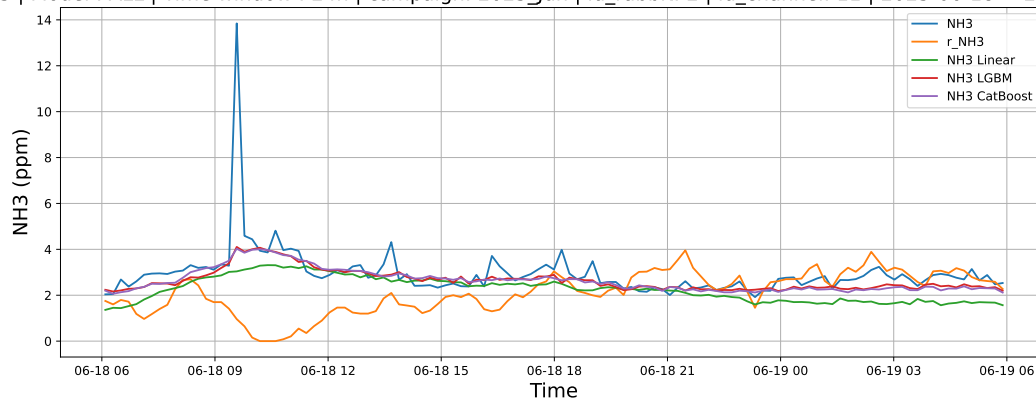
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

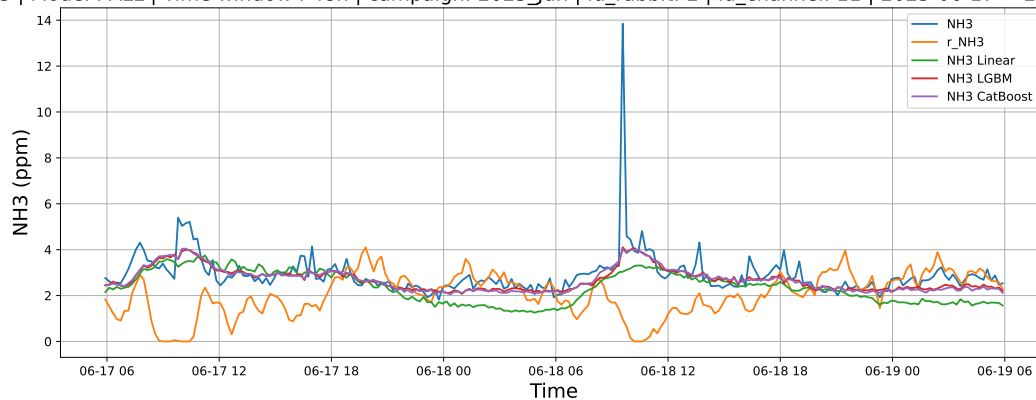
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



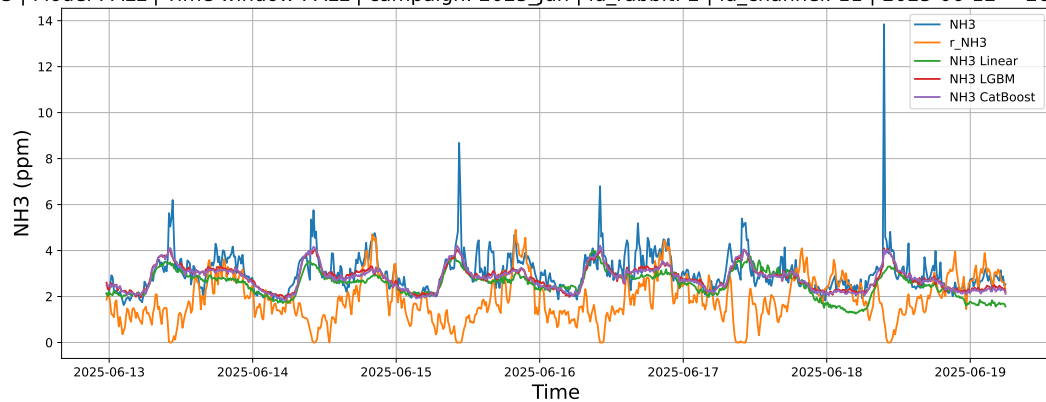
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

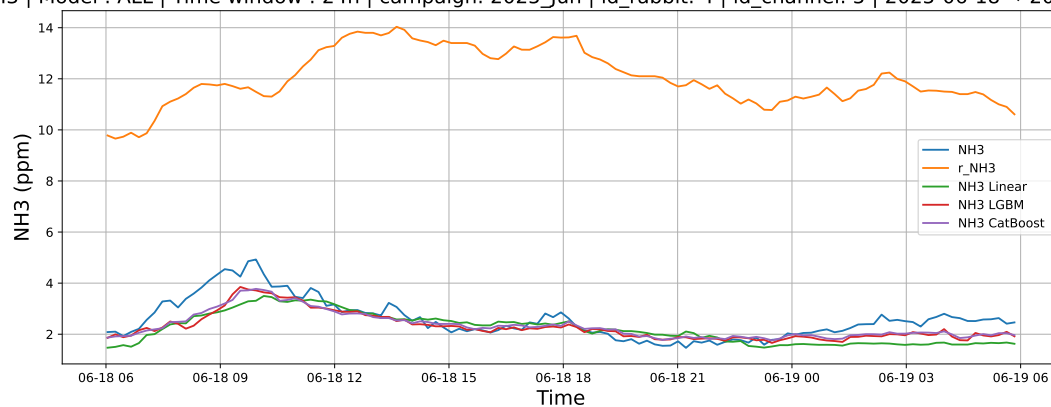
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

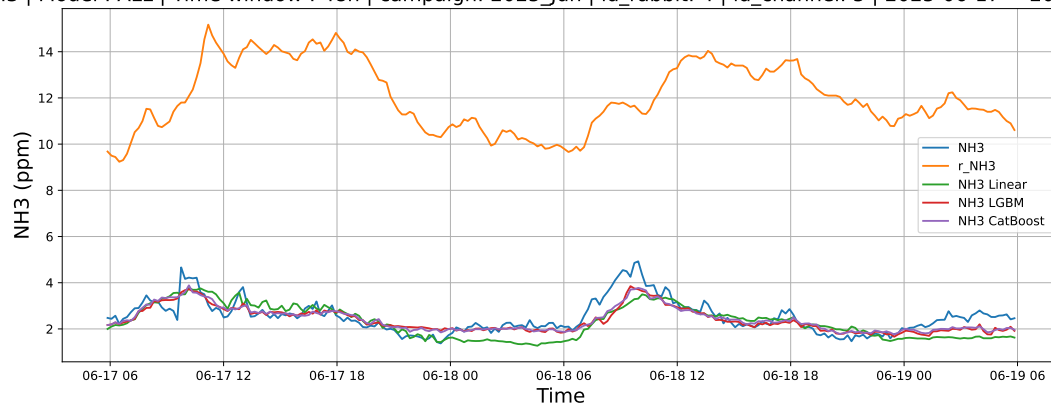
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



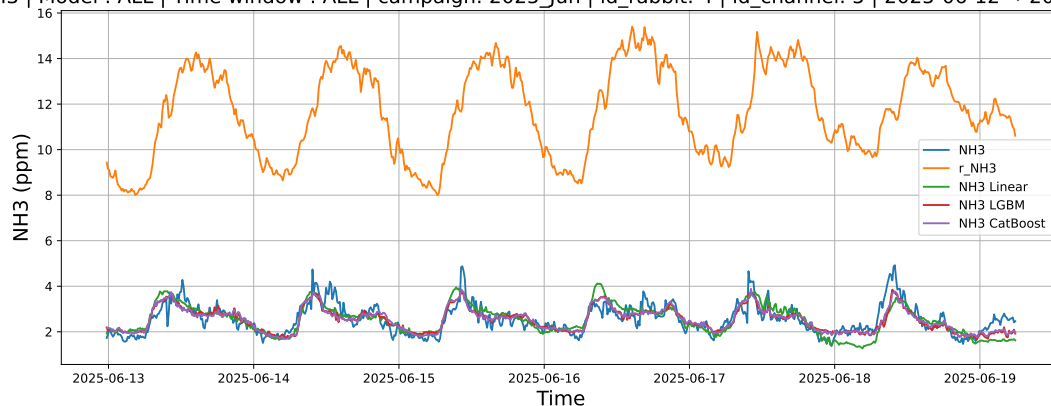
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

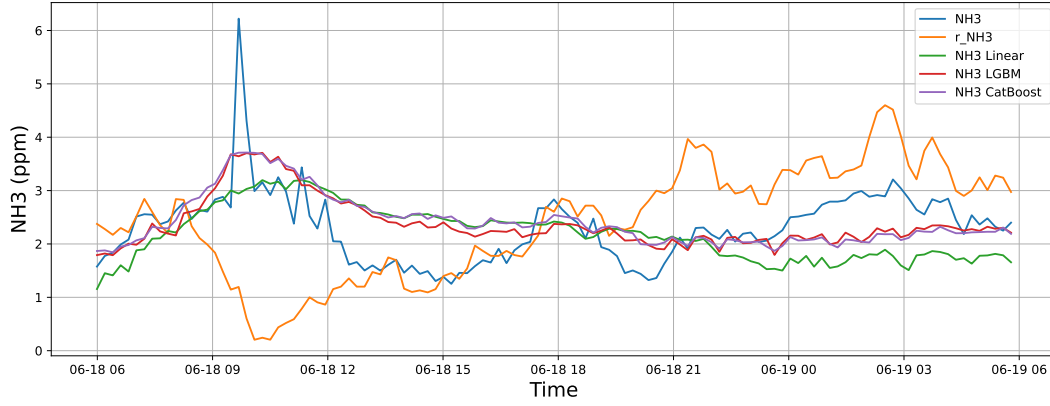
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

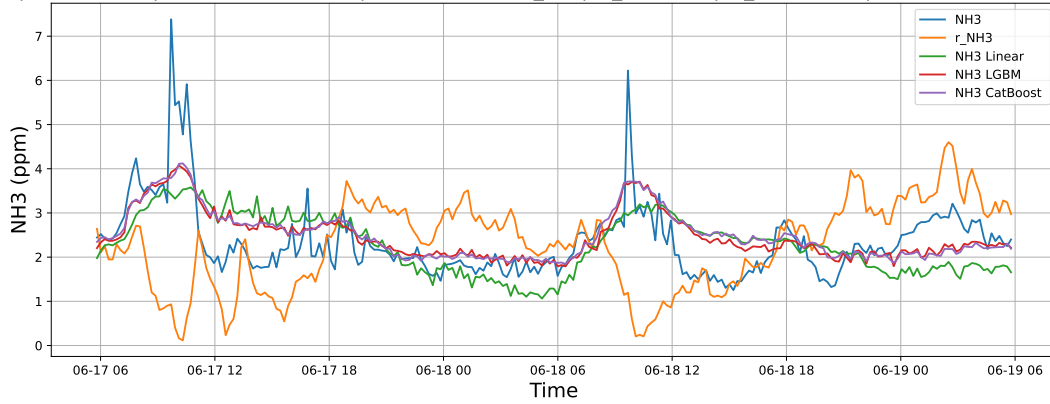
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



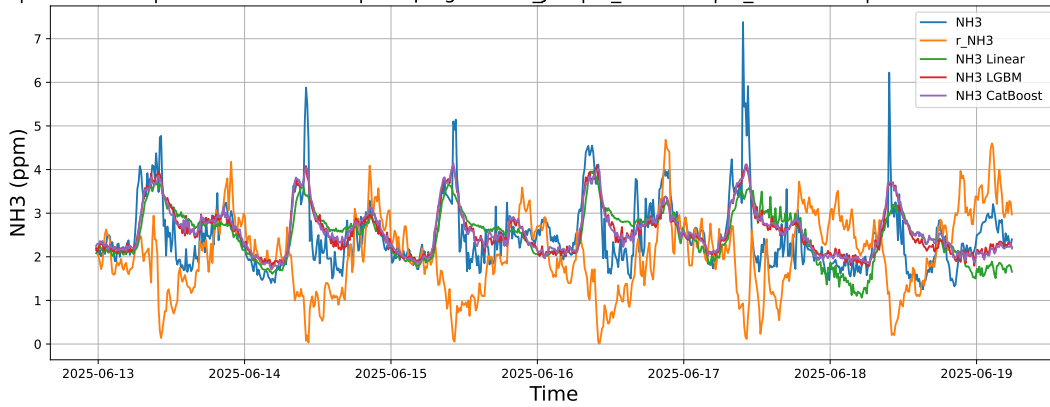
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

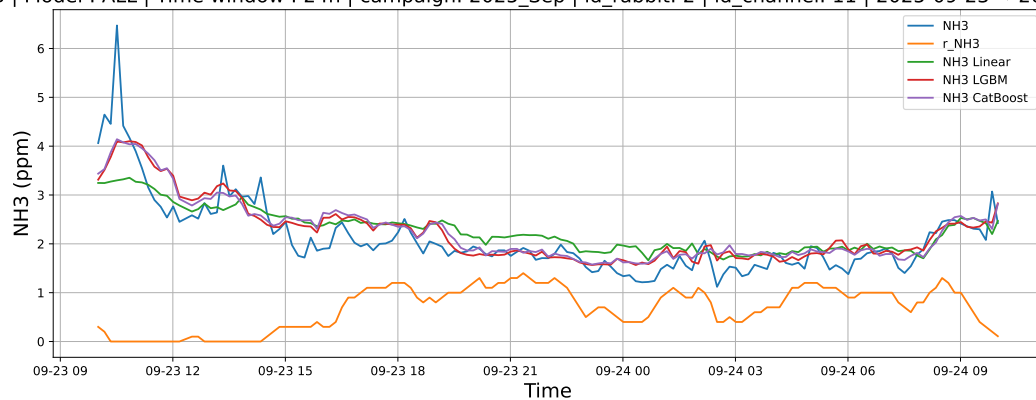
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

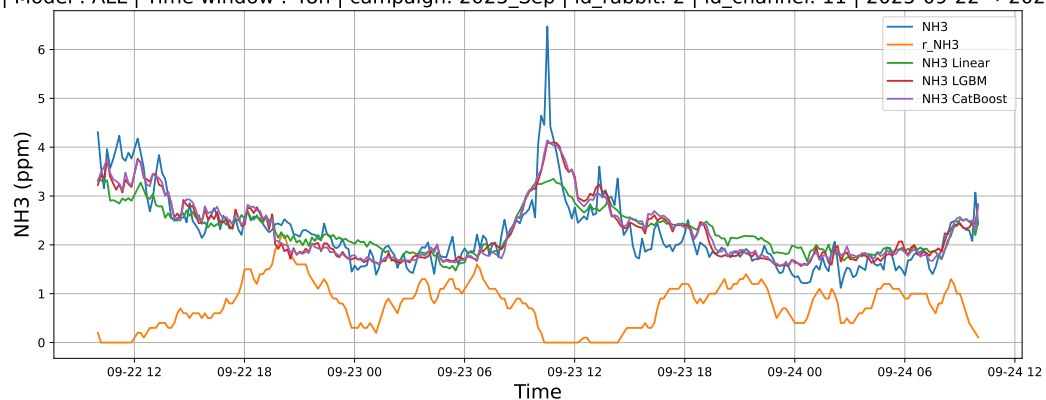
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



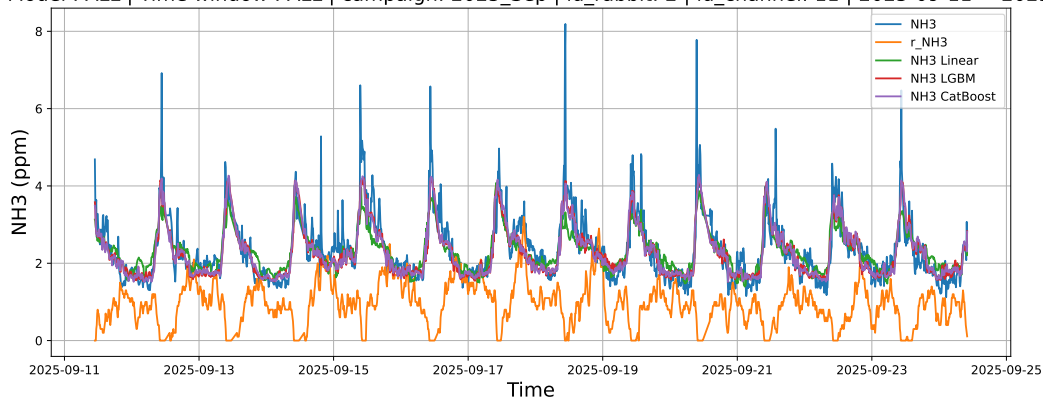
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

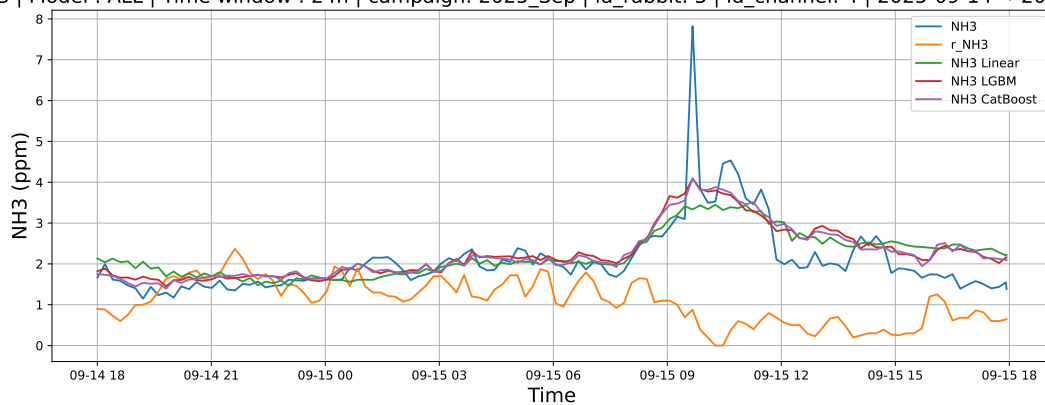
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

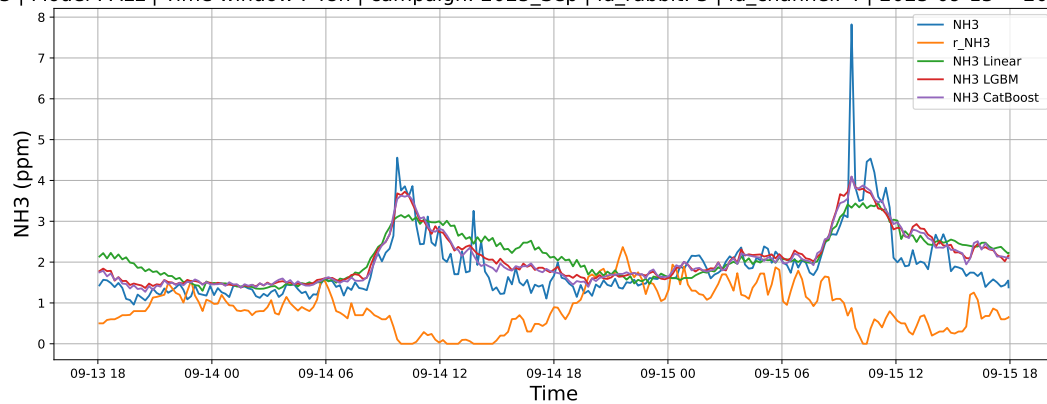
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



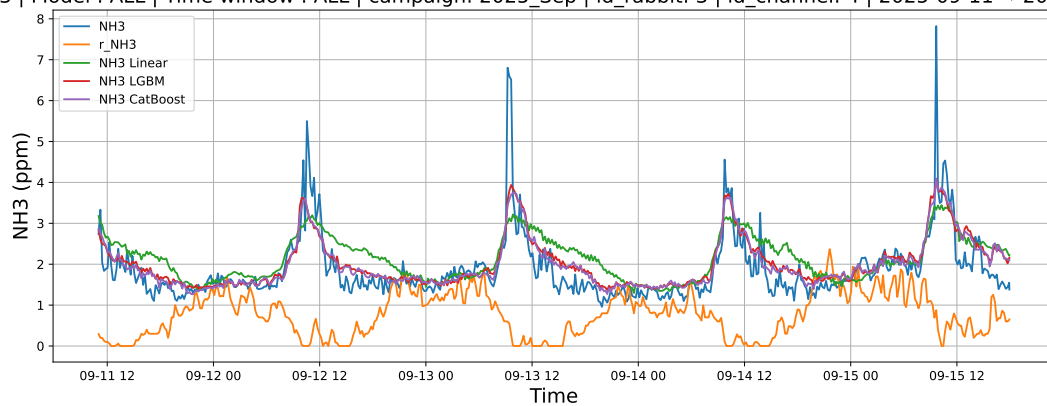
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

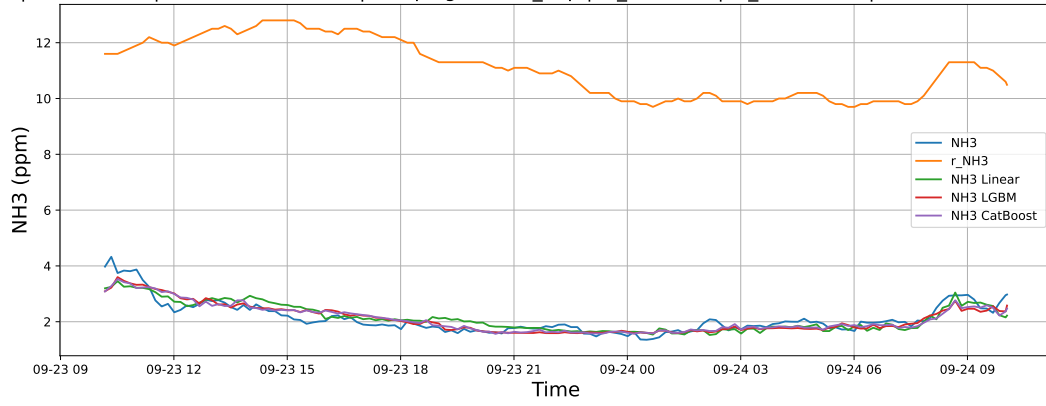
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

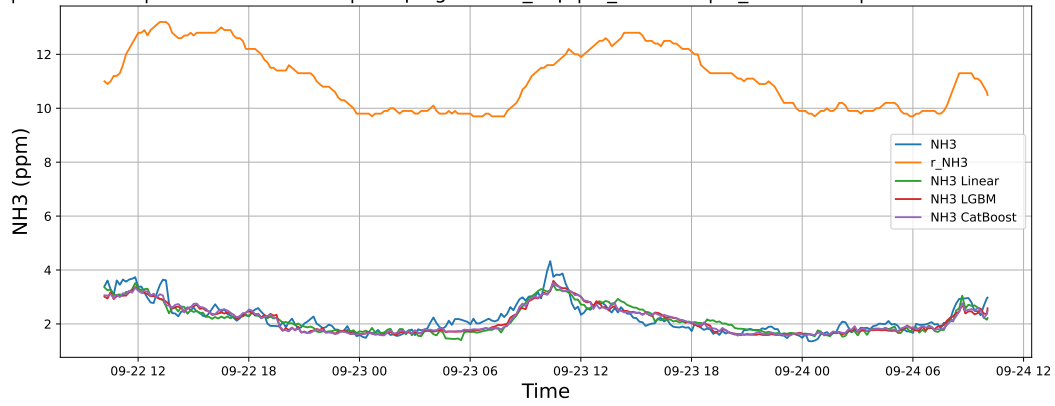
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



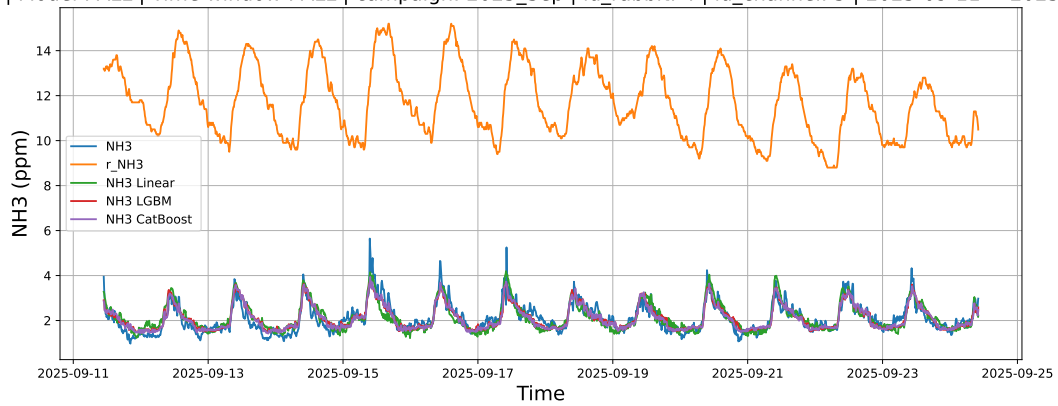
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

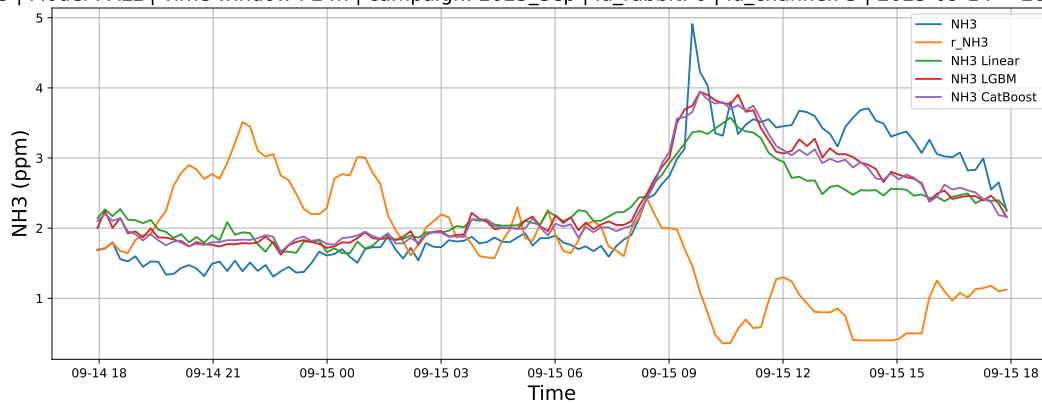
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

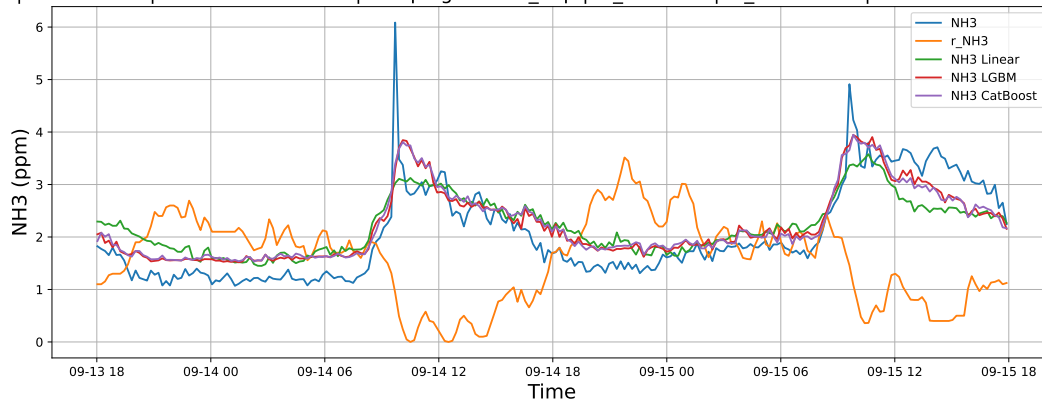
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

